Winding of AC Machines

Technical report

Winding patterns, EMF phasors,
magnetomotive force and harmonic analysis

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Chapter 1

Winding Theory

1.1 INTRODUCTION TO THE WINDING PROBLEM

The winding pattern generation problem involves distributing conductors across available stator slots in such a way that:

- Each phase is balanced (equal number of conductors and symmetrical distribution)
- The winding produces a sinusoidal MMF (magnetomotive force) distribution
- Maximum fundamental winding factor is achieved
- Harmonics are minimized (especially low-order harmonics)
- Mechanical symmetry is maintained for smooth torque production

The algorithm must handle:

- Any number of phases ($m = 2, 3, 4, 5, 6, \dots$)
- Any number of slots (Q)
- Any number of pole pairs (p)
- Integer slot windings (q is an integer)
- Fractional slot windings (q is a fraction)
- Single-layer windings (one coil side per slot)
- Double-layer windings (two coil sides per slot)

1.2 FUNDAMENTAL PARAMETERS AND CALCULATIONS

1.2.1 GIVEN PARAMETERS:

Let's define the input parameters:

- m = number of phases (typically 3, but can be 2, 4, 5, 6, etc.)
- Q = total number of slots in the stator
- p = number of pole pairs
- P = number of actual poles = $2p$

1.2.2 DERIVED PARAMETERS:

1. **SLOTS PER POLE PER PHASE (q):**

$$q = \frac{Q}{m \times P} = \frac{Q}{m \times 2p}$$

This is THE most critical parameter in winding design. It can be:

- An integer (e.g., $q = 2, 3, 4, \dots$) → INTEGER SLOT WINDING
- A fraction (e.g., $q = 5/6, 2/3, 7/8, \dots$) → FRACTIONAL SLOT WINDING

The value of q determines:

- Number of slots each phase occupies per pole
- Complexity of the winding pattern
- Winding symmetry and balance
- Harmonic content of the MMF

2. **MACHINE PERIODICITY (t):**

$$t = \gcd(Q, p)$$

where gcd is the greatest common divisor. This represents how many times the basic winding pattern repeats around the stator circumference.

Example: $Q = 36, p = 5 \rightarrow t = \gcd(36, 5) = 1$
$Q = 36, p = 3 \rightarrow t = \gcd(36, 3) = 3$

The periodicity affects:

- Pattern repetition
- Slot assignment complexity
- Symmetry requirements

3. **THEORETICAL POLE PITCH (τ_p):**

$$\tau_p = \frac{Q}{P} = \frac{Q}{2p} \quad \text{[in slots]}$$

This represents the theoretical span of one pole in terms of slots.

Example: $Q = 36, p = 5 \rightarrow \tau_p = 36/10 = 3.6$ slots

4. **COIL THROW (COIL SPAN) (y_q):**

   y_q = coil throw in slots (distance between coil sides)

   For FULL-PITCH winding: $y_q = \text{round}(\tau_p) = \text{round}(Q/2p)$
   For SHORT-PITCH (chorded): $y_q < \tau_p$
   For LONG-PITCH: $y_q > \tau_p$ (not recommended)

   In single-layer windings: y_q must be ODD
   In double-layer windings: y_q can be any value

5. **ELECTRICAL SLOT ANGLE (α_s):**

$$\alpha_s = \left(\frac{2\pi}{Q}\right) \times p \quad \text{[radians]}$$

   or

$$\alpha_s = \frac{360^\circ \times p}{Q} \quad \text{[degrees]}$$

   This is the electrical angle between adjacent slots.

   Example: $Q = 36, p = 5 \rightarrow \alpha_s = 360^\circ \times 5/36 = 50^\circ$

6. **PHASE ANGLE BETWEEN ADJACENT PHASORS (α_{ph}):**

$$\alpha_{ph} = \left(\frac{2\pi}{Q}\right) \times t \quad \text{[radians]}$$

   This represents the angle between adjacent spokes in the star of slots.

1.3 MACHINE PERIODICITY AND FEASIBILITY CHECKS

Before generating a winding, we must verify that a balanced winding is possible with the given parameters.

1.3.1 BALANCE CONDITION:

For a balanced m-phase winding, the following MUST be satisfied:

$$\frac{Q}{m \times t} = \text{integer}$$

where $t = \gcd(Q, p)$.

This ensures that each phase gets an equal number of slots per repeating period, allowing for perfect electrical and mechanical balance.

Example 1: $m = 3, Q = 36, p = 5, t = 1$
$Q/(m \times t) = 36/(3 \times 1) = 12$ INTEGER $\rightarrow$ BALANCED

Example 2: $m = 3, Q = 37, p = 5, t = 1$
$Q/(m \times t) = 37/(3 \times 1) = 12.33\ldots$ NOT INTEGER $\rightarrow$ NOT BALANCED

EVEN PHASE NUMBER ADDITIONAL CHECK:

When m is EVEN ($m = 2, 4, 6, \dots$), an additional check is needed:

$$\text{If } \gcd(m, 2) = 2 \text{ AND } \gcd(Q/(m \times t), 2) \neq 2, \text{ the winding is NOT feasible.}$$

This prevents imbalanced phase grouping in even-phase systems.

1.3.2 SINGLE LAYER FEASIBILITY:

Single-layer windings are more restrictive than double-layer windings. For a single-layer winding to be feasible:

1. Check if Q is even
   If Q is odd $\rightarrow$ Single layer NOT feasible
2. If Q is even, check Q/t:
   - **Case A:** If Q/t is ODD
     - If t is ODD $\rightarrow$ Single layer NOT feasible
     - If t is EVEN $\rightarrow$ Single layer FEASIBLE
   - **Case B:** If Q/t is EVEN
     - If $Q/(2t)$ is EVEN $\rightarrow$ Single layer FEASIBLE
     - Otherwise $\rightarrow$ Single layer FEASIBLE

The logic is:

- Q must be even (since we need paired slots)
- The pattern must allow alternating slot occupation
- Periodicity must align with slot pairing

Example: $Q = 36, p = 5, t = 1$
Q is even (36)
$Q/t = 36/1 = 36$ (even)
Single layer FEASIBLE

1.3.3 MINIMUM q REQUIREMENT:

Additionally, q must be reasonable:

$$q \geq 0.25$$

If $q < 0.25$, the winding becomes impractical (too few slots per phase).

1.4 THE STAR OF SLOTS THEORY

The "Star of Slots" is a visualization and computational technique where each stator slot is represented as a "spoke" (phasor) emanating from a center point, with angles determined by electrical position.

CREATING THE STAR:

Each slot 'i' (where $i = 1, 2, 3, \ldots, Q$) is represented by a spoke with:

$$\text{angle}_i = \alpha_s \times (i-1) + \varepsilon = \left[\left(\frac{2\pi}{Q}\right) \times p\right] \times (i-1) + \varepsilon$$

where:

- i is the slot number (1-indexed)
- α_s is the electrical slot angle
- ε (epsilon) is a small offset angle (explained in next section)

The angle represents the ELECTRICAL position of the slot, not mechanical.
Example for $Q = 36, p = 5$: $\alpha_s = (2\pi/36) \times 5 = 50°$ electrical

- Slot 1: angle $= 50° \times 0 + \varepsilon = 0° + \varepsilon$
- Slot 2: angle $= 50° \times 1 + \varepsilon = 50° + \varepsilon$
- Slot 3: angle $= 50° \times 2 + \varepsilon = 100° + \varepsilon$
- ...
- Slot 36: angle $= 50° \times 35 + \varepsilon = 1750° + \varepsilon = 310° + \varepsilon$ (mod 360°)

All angles are normalized to the range $[0, 2\pi)$ or $[0°, 360°)$.

1.4.1 ELECTRICAL vs MECHANICAL:

It's crucial to understand that the star of slots operates in the ELECTRICAL coordinate system:

- Mechanical angle: $\theta_{\text{mech}} = 2\pi \times (\text{slot_number}/Q)$
- Electrical angle: $\theta_{\text{elec}} = p \times \theta_{\text{mech}} = (2\pi \times p/Q) \times (\text{slot_number} - 1)$

This means that as we go mechanically around the stator once (360° mechanical), we go through p complete electrical cycles.
Example: With $p = 5$ pole pairs, going 360° mechanical = 1800° electrical (5 cycles)

1.5 EPSILON OFFSET CALCULATION (CRITICAL FOR SYMMETRY)

The epsilon (ε) offset is a CRITICAL small angular offset added to prevent spokes from landing exactly on sector boundaries, which would cause ambiguity in slot assignment.

1.5.1 WHY EPSILON IS NEEDED:

Without epsilon, some spokes might land exactly on the boundary between sectors, making it unclear which sector (and thus which phase) should claim that slot. This would lead to:

- Unbalanced phase distribution
- Asymmetric winding patterns
- Potential short circuits or missing conductors

1.5.2 EPSILON CALCULATION RULES:

The epsilon value depends on the number of phases (m) and the number of spokes per phase (spoke_per_phase):

$$\text{spoke_per_phase} = \frac{Q}{m \times t}$$

1. **CASE 1: ODD NUMBER OF PHASES ($m = 3, 5, 7, \ldots$):**
   - **Sub-case 1a:** If spoke_per_phase = 1 OR 2
   $$\varepsilon = \frac{\pi}{2m}$$
   - **Sub-case 1b:** If spoke_per_phase is EVEN (4, 6, 8, ...)
   $$\varepsilon = \frac{\alpha_{ph}}{2} = \frac{\pi}{m \times t}$$
   - **Sub-case 1c:** If spoke_per_phase is ODD (3, 5, 7, ...)
   $$\varepsilon = \frac{\alpha_{ph}}{4} = \frac{\pi}{2m \times t}$$
2. **CASE 2: EVEN NUMBER OF PHASES ($m = 2, 4, 6, \ldots$):**
   - **Sub-case 2a:** If spoke_per_phase = 1 OR 2
   $$\varepsilon = 0$$
   - **Sub-case 2b:** If (spoke_per_phase / 2) is EVEN
   $$\varepsilon = \frac{\alpha_{ph}}{4} = \frac{\pi}{2m \times t}$$
   - **Sub-case 2c:** Otherwise
   $$\varepsilon = \frac{\alpha_{ph}}{8} = \frac{\pi}{4m \times t}$$

1.5.3 EXAMPLE CALCULATION for $m = 3, Q = 36, p = 5$:

$t = \gcd(36, 5) = 1$
spoke_per_phase $= 36/(3 \times 1) = 12$

Since $m = 3$ (odd) and spoke_per_phase=12 (even):

$$\varepsilon = \frac{\alpha_{ph}}{2}$$

$$\alpha_{ph} = \left(\frac{2\pi}{36}\right) \times 1 = \frac{\pi}{18} \text{ radians}$$

$$\varepsilon = \frac{\pi/18}{2} = \frac{\pi}{36} \text{ radians}$$

$$\varepsilon \approx 5^\circ \text{ electrical}$$

This small 5° offset ensures no spoke lands on a sector boundary.

1.5.4 PHYSICAL INTERPRETATION:

Think of epsilon as a "rotation" of the entire star to ensure clean sector boundaries. It's like slightly rotating a key to align it perfectly with a lock. The value is chosen to be small enough not to affect the fundamental winding behavior, but large enough to ensure unambiguous slot assignment.

1.6 SECTOR GENERATION AND SLOT ASSIGNMENT

Once we have the star of slots (all spokes with their angles), we divide the 360° electrical circle into SECTORS, with each phase getting one positive and one negative sector per electrical cycle.

1.6.1 SECTOR BASICS:

For m phases, we create:

- m POSITIVE sectors (for positive current/MMF contribution)
- m NEGATIVE sectors (for negative current/MMF contribution)

Total $= 2m$ sectors per electrical cycle.
Each sector spans: $2\pi/(2m) = \pi/m$ radians $= 180^\circ/m$

1.6.2 SECTOR ANGLE FORMULAS:

The sectors are positioned with an offset to align with traditional phase conventions (Phase A aligned with vertical/top position).

For PHASE i (where $i = 0, 1, 2, \ldots, m-1$):

1. **ODD NUMBER OF PHASES ($m = 3, 5, 7, \dots$):**

   Positive sector for phase i:

$$\text{start_angle} = \text{offset} + \frac{\pi}{2} + \left(\frac{\pi}{m}\right) \times (2i - 0.5)$$

$$\text{end_angle} = \text{offset} + \frac{\pi}{2} + \left(\frac{\pi}{m}\right) \times (2i + 0.5)$$

   Negative sector for phase i:

$$\text{start_angle} = \text{offset} + \frac{\pi}{2} + \left(\frac{\pi}{m}\right) \times (2i - 0.5) + \pi$$

$$\text{end_angle} = \text{offset} + \frac{\pi}{2} + \left(\frac{\pi}{m}\right) \times (2i + 0.5) + \pi$$

2. **EVEN NUMBER OF PHASES ($m = 2, 4, 6, \dots$):**

   Positive sector for phase i:

$$\text{start_angle} = \text{offset} + \frac{\pi}{2} + \left(\frac{\pi}{m}\right) \times (i - 0.5)$$

$$\text{end_angle} = \text{offset} + \frac{\pi}{2} + \left(\frac{\pi}{m}\right) \times (i + 0.5)$$

   Negative sector for phase i:

$$\text{start_angle} = \text{offset} + \frac{\pi}{2} + \left(\frac{\pi}{m}\right) \times (i - 0.5) + \pi$$

$$\text{end_angle} = \text{offset} + \frac{\pi}{2} + \left(\frac{\pi}{m}\right) \times (i + 0.5) + \pi$$

   where 'offset' is typically 0 (can be adjusted for special phase alignments).

1.6.3 EXAMPLE for 3-PHASE ($m = 3$):

Each sector spans: $180°/3 = 60°$

Phase A ($i = 0$):

- Positive sector: $[90° + 60° \times (0 - 0.5), 90° + 60° \times (0 + 0.5)] = [60°, 120°]$
- Negative sector: $[60° + 180°, 120° + 180°] = [240°, 300°]$

Phase B ($i = 1$):

- Positive sector: $[90° + 60° \times (2 - 0.5), 90° + 60° \times (2 + 0.5)] = [180°, 240°]$
- Negative sector: $[180° + 180°, 240° + 180°] = [360° \equiv 0°, 60°]$

Phase C ($i = 2$):

- Positive sector: $[90° + 60° \times (4 - 0.5), 90° + 60° \times (4 + 0.5)] = [300°, 360° \equiv 0°]$
- Negative sector: $[300° + 180°, 360° + 180°] = [480° \equiv 120°, 180°]$

Note: Angles are wrapped to $[0°, 360°)$ range.

1.6.4 SECTOR WRAP-AROUND HANDLING:

Some sectors will "wrap around" the $0°/360°$ boundary. For example, a sector might span from $330°$ to $30°$. The algorithm must handle this correctly when checking if a spoke belongs to a sector.

To check if angle θ is inside sector $[\text{start}, \text{end}]$:

- IF start $>$ end: (wrap-around case)
  θ is inside IF ($\theta >$ start) OR ($\theta <$ end)
- ELSE: (normal case)
  θ is inside IF ($\theta >$ start) AND ($\theta <$ end)

1.6.5 SLOT ASSIGNMENT TO SECTORS:

After creating all sectors, we iterate through each spoke in the star and assign it to the appropriate sector:

For each spoke with angle θ:

- For each positive sector:
  - If θ is inside this sector:
    - Add spoke's slot number to this sector's positive slot list
- For each negative sector:
  - If θ is inside this sector:
    - Add spoke's slot number to this sector's negative slot list

After this process, each sector knows which slots belong to it.
Example for $m = 3, Q = 36, p = 5$:

- Phase A positive sector $[60°, 120°]$ might contain slots: $[2, 3, 14, 15, 26, 27]$
- Phase A negative sector $[240°, 300°]$ might contain slots: $[8, 9, 20, 21, 32, 33]$
- (exact slots depend on epsilon and specific angles)

1.7 COIL GENERATION AND WINDING CONSTRUCTION

Now that we know which slots belong to each phase (from sector assignment), we generate COILS for each phase.

1.7.1 COIL STRUCTURE:

A coil consists of:

- begin: Starting slot number (1-indexed)
- end: Ending slot number (1-indexed)
- nc: Number of conductors (positive or negative)

The coil span (distance from begin to end) is determined by the coil throw y_q.

1.7.2 DOUBLE LAYER WINDING COIL GENERATION:

For DOUBLE LAYER winding (most common):
For each phase i:

1. Process POSITIVE sector slots:
   For each slot s in positive_sector[i]:
   - begin = s
   - end = $s + y_q$
   - IF end > Q: end = end $-Q$ (wrap around)
   - Create Coil(begin=begin, end=end, nc=+1)
   - Add to phase i winding
2. Process NEGATIVE sector slots:
   For each slot s in negative_sector[i]:
   - begin = s
   - end = $s + y_q$
   - IF end > Q: end = end $-Q$ (wrap around)
   - Create Coil(begin=begin, end=end, nc=-1)
   - Add to phase i winding

The nc (number of conductors) is:

- +1 for positive sector coils (current in one direction)
- -1 for negative sector coils (current in opposite direction)

1.7.3 COIL INTERPRETATION:

In a double-layer winding:

- The "begin" slot contains the TOP layer of the coil
- The "end" slot contains the BOTTOM layer of the coil
- All slots have two layers: top and bottom

When nc = +1: Current flows INTO the page at the top layer (begin slot)
Current flows OUT of the page at the bottom layer (end slot)
When nc = -1: Reverse polarity (opposite current direction)

1.7.4 SINGLE LAYER WINDING COIL GENERATION:

For SINGLE LAYER winding:
The process is similar, but with restrictions:

- Only ODD-numbered slots can be used as starting points
- y_q must be ODD
- Each slot contains only ONE coil side

For each phase i:

1. Process POSITIVE sector slots:
   For each slot s in positive_sector[i]:
   - IF s is EVEN: SKIP this slot (single layer only uses odd slots)
   - begin = s
   - end = $s + y_q$
   - IF end $> Q$: end = end $-Q$
   - Create Coil(begin=begin, end=end, nc=+1)
   - Add to phase i winding
2. Process NEGATIVE sector slots:
   (Same logic with even slots skipped)

1.7.5 COIL THROW (y_q) DETERMINATION:

If not specified by the user:

$$y_q = \text{theoretical pole pitch} = \text{round}\left(\frac{Q}{2p}\right)$$

For single layer: If y_q is even, make it odd: $y_q = y_q - 1$
For double layer: y_q can be any value, but typically:

- Full pitch: $y_q = \text{round}(Q/2p)$
- Short pitch: $y_q <$ theoretical pole pitch (common: 80-90% of pole pitch)

1.7.6 EXAMPLE for $m = 3, Q = 36, p = 5$:

Theoretical pole pitch: $\tau_p = 36/10 = 3.6 \approx 4$ slots
For double layer with $y_q = 4$:
Phase A might have coils like:

- Coil(begin=2, end=6, nc=+1) [slot 2 top layer, slot 6 bottom layer, positive]
- Coil(begin=3, end=7, nc=+1) [slot 3 top layer, slot 7 bottom layer, positive]
- ...
- Coil(begin=8, end=12, nc=-1) [slot 8 top layer, slot 12 bottom layer, negative]
- ...

1.8 WINDING FACTORS CALCULATION

Winding factors quantify how effectively the winding produces a sinusoidal MMF distribution. There are three main factors:

1.8.1 DISTRIBUTION FACTOR (k_d):

Accounts for the distribution of coils across multiple slots per pole per phase.

For the fundamental ($\nu = 1$):

$$k_d = \frac{\sin(q \times \alpha/2)}{q \times \sin(\alpha/2)}$$

where:

- q = slots per pole per phase (can be fractional)
- α = electrical slot angle = $\pi \times P/Q = 2\pi \times p/Q$

For fractional slot windings, use the fractional value of q directly.

Example: $q = 5/6, \alpha = 50° = 0.873$ rad

$$k_d = \frac{\sin(5/6 \times 0.873/2)}{(5/6) \times \sin(0.873/2)} = \frac{\sin(0.364)}{0.833 \times \sin(0.436)} = \frac{0.356}{0.833 \times 0.422} = \frac{0.356}{0.352} \approx 1.011 \text{ (normalized to } \leq 1.$$

Physical meaning: If $k_d = 1.0$, coils are perfectly concentrated (no distribution).
Typical values: 0.90 to 0.96 for distributed windings.

1.8.2 PITCH FACTOR (CHORDING FACTOR) (k_p):

Accounts for the coil not spanning a full pole pitch (short-pitching or chorded).

$$k_p = \sin(\gamma/2)$$

where γ is the coil pitch in electrical radians:

$$\gamma = y_q \times \alpha = y_q \times \left(\frac{2\pi \times p}{Q}\right)$$

Example: $y_q = 4$ slots, $\alpha = 50° = 0.873$ rad

$$\gamma = 4 \times 0.873 = 3.49 \text{ rad} = 200° \text{ electrical}$$

$$k_p = \sin(200°/2) = \sin(100°) = 0.9848$$

Physical meaning: If $k_p = 1.0$, coil is full-pitch ($\gamma = 180° = \pi$ rad).
If $k_p < 1.0$, coil is short-pitched (chorded).
Short pitching reduces fundamental but eliminates certain harmonics.

Typical values:

- Full pitch: $k_p = 1.0$
- 80% pitch: $k_p \approx 0.94$
- 5/6 pitch: $k_p \approx 0.966$

1.8.3 SKEW FACTOR (k_s):

Accounts for axial skewing of the rotor or stator to reduce cogging and noise.

$$k_s = \frac{\sin(\sigma/2)}{\sigma/2}$$

where σ is the skew angle in electrical radians.
For NO skew: $\sigma = 0$, and $k_s = 1.0$ (limit as $\sigma \to 0$)
For one slot skew: $\sigma = \alpha$ (one slot angle)

$$k_s = \frac{\sin(\alpha/2)}{\alpha/2}$$

Example: $\alpha = 50° = 0.873$ rad, $\sigma = 0.873$ rad

$$k_s = \frac{\sin(0.873/2)}{0.873/2} = \frac{\sin(0.436)}{0.436} = \frac{0.422}{0.436} \approx 0.968$$

Most designs assume no skew initially: $k_s = 1.0$

1.8.4 TOTAL WINDING FACTOR (k_w):

$$k_w = k_d \times k_p \times k_s$$

This is the overall effectiveness of the winding.
Example: $k_d = 0.956, k_p = 0.985, k_s = 1.0$

$$k_w = 0.956 \times 0.985 \times 1.0 = 0.942$$

Physical meaning: The actual MMF produced is 94.2% of the theoretical maximum if all coils were concentrated and full-pitch.

Typical values for good designs: $k_w = 0.85$ to 0.96

1.8.5 HARMONICS CONSIDERATION:

Each harmonic order ($\nu = 1, 3, 5, 7, 9, \dots$) has its own set of winding factors:

$$k_{d_\nu} = \frac{\sin(\nu \times q \times \alpha/2)}{q \times \sin(\nu \times \alpha/2)}$$

$$k_{p_\nu} = \sin(\nu \times \gamma/2)$$

$$k_{w_\nu} = k_{d_\nu} \times k_{p_\nu} \times k_{s_\nu}$$

Short-pitching can ELIMINATE specific harmonics by choosing γ such that:

$$\sin(\nu \times \gamma/2) = 0$$

$$\Rightarrow \nu \times \gamma/2 = n \times \pi$$

$$\Rightarrow \gamma = \frac{2n \times \pi}{\nu}$$

Example: To eliminate 5th harmonic ($\nu = 5$), choose $\gamma = 2\pi/5 = 144°$ electrical.

1.9 DOUBLE LAYER VS SINGLE LAYER IMPLEMENTATION

1.9.1 DOUBLE LAYER WINDING:

Structure:

- Each slot has TWO layers: top and bottom
- Top layer: where coils BEGIN
- Bottom layer: where coils END
- Allows ANY coil span (y_q can be even or odd)
- More flexible and more common in modern machines

Matrix Representation: $2m \times Q$ matrix, where:

- Rows 0 to $m-1$: Top layer for phases 1 to m
- Rows m to $2m-1$: Bottom layer for phases 1 to m
- Each element indicates phase and polarity in that slot

Example for 3-phase:

- Row 0: Phase A top layer $[+1, 0, 0, +1, 0, 0, \dots]$
- Row 1: Phase B top layer $[0, 0, +1, 0, 0, +1, \dots]$
- Row 2: Phase C top layer $[0, +1, 0, 0, +1, 0, \dots]$
- Row 3: Phase A bottom layer $[-1, 0, 0, -1, 0, 0, \dots]$
- Row 4: Phase B bottom layer $[0, 0, -1, 0, 0, -1, \dots]$
- Row 5: Phase C bottom layer $[0, -1, 0, 0, -1, 0, \dots]$

Construction:

- Top layer is occupied by coil START (begin slot)
- Bottom layer is occupied by coil END (end slot)
- Allows overlapping coils
- Better slot utilization
- Easier to manufacture with pre-formed coils

Advantages:

+ Better slot fill factor (up to 50%)
+ Shorter end windings

+ Better cooling

+ Better harmonic reduction

+ More design flexibility

Disadvantages:

- More complex to wind

- Requires precise layer insulation

- Slightly more copper (two layers)

SINGLE LAYER WINDING:

Structure:

- Each slot has ONE layer only
- Only odd-numbered slots are used (or only even, depending on convention)
- y_q must be ODD
- More restrictive design rules

Matrix Representation: $m \times Q$ matrix, where:

- Rows 0 to $m - 1$: Single layer for phases 1 to m
- Each element indicates phase and polarity in that slot
- Many zeros (empty slots or unused layer)

Example for 3-phase:

- Row 0: Phase A $[+1, 0, -1, 0, +1, 0, -1, 0, \dots]$
- Row 1: Phase B $[0, -1, 0, +1, 0, -1, 0, +1, \dots]$
- Row 2: Phase C $[-1, 0, +1, 0, -1, 0, +1, 0, \dots]$

Construction:

- Only one coil side per slot
- Coils do not overlap
- Simpler physical construction
- Often uses lap winding or wave winding

Advantages:

+ Simpler to manufacture

+ Better phase insulation

+ Easier to repair
+ Higher reliability
+ Stronger fundamental (less distribution)

Disadvantages:

- Lower slot utilization (only ∼40%)
- Longer end windings (odd span)
- More harmonic content
- More restrictive design space
- Not feasible for all Q, p combinations

1.9.2 WHEN TO USE EACH:

Use DOUBLE LAYER when:

- High power density needed
- Fractional slot design
- Flexible coil pitch adjustment needed
- Standard modern machine design

Use SINGLE LAYER when:

- Simplicity and reliability are critical
- Low-power applications
- Easy maintenance required
- Cost reduction is priority
- q is small (1 or 2)

1.10 COMPLETE WORKED EXAMPLE: 3 PHASES, 36 SLOTS, 5 POLE PAIRS

Let's work through the COMPLETE algorithm for a real machine:

1.10.1 GIVEN PARAMETERS:

- $m = 3$ phases
- $Q = 36$ slots
- $p = 5$ pole pairs
- $P = 2p = 10$ poles
- Winding type: Double layer (we'll also discuss single layer)

1.10.2 STEP 1: CALCULATE FUNDAMENTAL PARAMETERS

1. Slots per pole per phase:

$$q = \frac{Q}{m \times P} = \frac{36}{3 \times 10} = \frac{36}{30} = \frac{6}{5} = 1.2$$

   ⇒ This is a FRACTIONAL SLOT WINDING (q is not an integer)

2. Machine periodicity:

$$t = \gcd(Q, p) = \gcd(36, 5)$$

   Using Euclidean algorithm: $\gcd(36, 5) = \gcd(5, 36 \mod 5) = \gcd(5, 1) = \gcd(1, 0) = 1$

$$t = 1$$

   ⇒ Pattern does NOT repeat (unique around circumference)

3. Theoretical pole pitch:

$$\tau_p = \frac{Q}{P} = \frac{36}{10} = 3.6 \text{ slots}$$

   ⇒ Not an integer! Coil span must be chosen as 3 or 4 slots

4. Electrical slot angle:

$$\alpha_s = \frac{360° \times p}{Q} = \frac{360° \times 5}{36} = 50° \text{ electrical}$$

   or in radians:

$$\alpha_s = \frac{2\pi \times 5}{36} = \frac{10\pi}{36} = \frac{5\pi}{18} \approx 0.8727 \text{ radians}$$

5. Phase angle between adjacent phasors:

$$\alpha_{ph} = \frac{2\pi \times t}{Q} = \frac{2\pi \times 1}{36} = \frac{\pi}{18} \approx 0.1745 \text{ radians} = 10° \text{ electrical}$$

1.10.3 STEP 2: CHECK FEASIBILITY

Balance condition:

$$\frac{Q}{m \times t} = \frac{36}{3 \times 1} = 12 \quad \text{INTEGER}$$

⇒ Balanced winding is FEASIBLE

Check q minimum:

$$q = 1.2 \geq 0.25$$

⇒ Practical design

Single layer feasibility:

- $Q = 36$ (even)
- $Q/t = 36/1 = 36$ (even)
- $Q/(2t) = 36/2 = 18$ (even)

⇒ Single layer is FEASIBLE (though double layer is better for this case)

1.10.4 STEP 3: CALCULATE EPSILON OFFSET

Spokes per phase per period:

$$\text{spoke_per_phase} = \frac{Q}{m \times t} = \frac{36}{3 \times 1} = 12$$

Since $m = 3$ (odd) and spoke_per_phase $= 12$ (even): Use formula: $\varepsilon = \alpha_{ph}/2$

$$\varepsilon = \frac{\pi/18}{2} = \frac{\pi}{36} \text{ radians} = 5^\circ \text{ electrical}$$

1.10.5 STEP 4: CREATE STAR OF SLOTS

Generate 36 spokes with angles:

Spoke (slot) i has angle: $\theta_i = \alpha_s \times (i-1) + \varepsilon = 50^\circ \times (i-1) + 5^\circ$

Let's calculate angles for all 36 slots:

- Slot 1: $\theta = 50^\circ \times 0 + 5^\circ = 5^\circ$
- Slot 2: $\theta = 50^\circ \times 1 + 5^\circ = 55^\circ$
- Slot 3: $\theta = 50^\circ \times 2 + 5^\circ = 105^\circ$
- Slot 4: $\theta = 50^\circ \times 3 + 5^\circ = 155^\circ$
- Slot 5: $\theta = 50^\circ \times 4 + 5^\circ = 205^\circ$
- Slot 6: $\theta = 50^\circ \times 5 + 5^\circ = 255^\circ$
- Slot 7: $\theta = 50^\circ \times 6 + 5^\circ = 305^\circ$
- Slot 8: $\theta = 50^\circ \times 7 + 5^\circ = 355^\circ$
- Slot 9: $\theta = 50^\circ \times 8 + 5^\circ = 405^\circ = 45^\circ \pmod{360^\circ}$
- Slot 10: $\theta = 50^\circ \times 9 + 5^\circ = 455^\circ = 95^\circ$
- Slot 11: $\theta = 50^\circ \times 10 + 5^\circ = 505^\circ = 145^\circ$
- Slot 12: $\theta = 50^\circ \times 11 + 5^\circ = 555^\circ = 195^\circ$
- Slot 13: $\theta = 50^\circ \times 12 + 5^\circ = 605^\circ = 245^\circ$
- Slot 14: $\theta = 50^\circ \times 13 + 5^\circ = 655^\circ = 295^\circ$
- Slot 15: $\theta = 50^\circ \times 14 + 5^\circ = 705^\circ = 345^\circ$
- Slot 16: $\theta = 50^\circ \times 15 + 5^\circ = 755^\circ = 35^\circ$
- Slot 17: $\theta = 50^\circ \times 16 + 5^\circ = 805^\circ = 85^\circ$
- Slot 18: $\theta = 50^\circ \times 17 + 5^\circ = 855^\circ = 135^\circ$
- Slot 19: $\theta = 50^\circ \times 18 + 5^\circ = 905^\circ = 185^\circ$
- Slot 20: $\theta = 50^\circ \times 19 + 5^\circ = 955^\circ = 235^\circ$
- Slot 21: $\theta = 50^\circ \times 20 + 5^\circ = 1005^\circ = 285^\circ$
- Slot 22: $\theta = 50^\circ \times 21 + 5^\circ = 1055^\circ = 335^\circ$
- Slot 23: $\theta = 50^\circ \times 22 + 5^\circ = 1105^\circ = 25^\circ$
- Slot 24: $\theta = 50^\circ \times 23 + 5^\circ = 1155^\circ = 75^\circ$
- Slot 25: $\theta = 50^\circ \times 24 + 5^\circ = 1205^\circ = 125^\circ$
- Slot 26: $\theta = 50^\circ \times 25 + 5^\circ = 1255^\circ = 175^\circ$
- Slot 27: $\theta = 50^\circ \times 26 + 5^\circ = 1305^\circ = 225^\circ$
- Slot 28: $\theta = 50^\circ \times 27 + 5^\circ = 1355^\circ = 275^\circ$
- Slot 29: $\theta = 50^\circ \times 28 + 5^\circ = 1405^\circ = 325^\circ$

- Slot 30: $\theta = 50° \times 29 + 5° = 1455° = 15°$
- Slot 31: $\theta = 50° \times 30 + 5° = 1505° = 65°$
- Slot 32: $\theta = 50° \times 31 + 5° = 1555° = 115°$
- Slot 33: $\theta = 50° \times 32 + 5° = 1605° = 165°$
- Slot 34: $\theta = 50° \times 33 + 5° = 1655° = 215°$
- Slot 35: $\theta = 50° \times 34 + 5° = 1705° = 265°$
- Slot 36: $\theta = 50° \times 35 + 5° = 1755° = 315°$

1.10.6 STEP 5: CREATE SECTORS FOR 3 PHASES

Each sector spans $180°/3 = 60°$

Phase A ($i = 0$):

- Positive sector:

$$\begin{aligned} \text{start} &= 90° + 60° \times (2 \times 0 - 0.5) \\ &= 90° + 60° \times (-0.5) \\ &= 90° - 30° = 60° \\ \text{end} &= 90° + 60° \times (2 \times 0 + 0.5) \\ &= 90° + 60° \times (0.5) \\ &= 90° + 30° = 120° \end{aligned}$$

  Sector A+: $[60°, 120°]$

- Negative sector:

$$\begin{aligned} \text{start} &= 60° + 180° = 240° \\ \text{end} &= 120° + 180° = 300° \end{aligned}$$

  Sector A-: $[240°, 300°]$

Phase B ($i = 1$):

- Positive sector:

$$\begin{aligned} \text{start} &= 90° + 60° \times (2 \times 1 - 0.5) \\ &= 90° + 60° \times 1.5 \\ &= 90° + 90° = 180° \\ \text{end} &= 90° + 60° \times (2 \times 1 + 0.5) \\ &= 90° + 60° \times 2.5 \\ &= 90° + 150° = 240° \end{aligned}$$

  Sector B+: $[180°, 240°]$

- Negative sector:

$$\begin{aligned} \text{start} &= 180° + 180° = 360° \equiv 0° \\ \text{end} &= 240° + 180° = 420° \equiv 60° \end{aligned}$$

  Sector B-: $[0°, 60°]$ (wraps around)

Phase C ($i = 2$):

- Positive sector:

$$\begin{aligned}\text{start} &= 90^\circ + 60^\circ \times (2 \times 2 - 0.5) \\ &= 90^\circ + 60^\circ \times 3.5 \\ &= 90^\circ + 210^\circ = 300^\circ \\ \text{end} &= 90^\circ + 60^\circ \times (2 \times 2 + 0.5) \\ &= 90^\circ + 60^\circ \times 4.5 \\ &= 90^\circ + 270^\circ = 360^\circ \equiv 0^\circ\end{aligned}$$

  Sector C+: $[300^\circ, 360^\circ \equiv 0^\circ]$ (wraps around)

- Negative sector:

$$\begin{aligned}\text{start} &= 300^\circ + 180^\circ = 480^\circ \equiv 120^\circ \\ \text{end} &= 360^\circ + 180^\circ = 540^\circ \equiv 180^\circ\end{aligned}$$

  Sector C-: $[120^\circ, 180^\circ]$

Summary of sectors:

- Phase A+: $[60^\circ, 120^\circ]$
- Phase A-: $[240^\circ, 300^\circ]$
- Phase B+: $[180^\circ, 240^\circ]$
- Phase B-: $[0^\circ, 60^\circ]$ (wrap)
- Phase C+: $[300^\circ, 360^\circ \equiv 0^\circ]$ (wrap)
- Phase C-: $[120^\circ, 180^\circ]$

1.10.7 STEP 6: ASSIGN SLOTS TO SECTORS

Now we check each slot angle to see which sector it belongs to:

- **Phase A+** $[60^\circ, 120^\circ]$:
  - Slot 31: 65°
  - Slot 24: 75°
  - Slot 17: 85°
  - Slot 10: 95°
  - Slot 3: 105°
  - Slot 32: 115°
  - Total: 6 slots
- **Phase A-** $[240^\circ, 300^\circ]$:
  - Slot 13: 245°
  - Slot 6: 255°
  - Slot 35: 265°
  - Slot 28: 275°
  - Slot 21: 285°
  - Slot 14: 295°
  - Total: 6 slots

- **Phase B+** $[180°, 240°]$:
  - Slot 19: 185°
  - Slot 12: 195°
  - Slot 5: 205°
  - Slot 34: 215°
  - Slot 27: 225°
  - Slot 20: 235°
  - Total: 6 slots
- **Phase C+** $[300°, 360° \equiv 0°]$ (wrap): Angles $> 300°$ OR $< 0°$
  - Slot 7: 305°
  - Slot 36: 315°
  - Slot 29: 325°
  - Slot 22: 335°
  - Slot 15: 345°
  - Slot 8: 355°
  - Total: 6 slots
- **Phase B-** $[0°, 60°]$ (wrap): Angles $> 0°$ AND $< 60°$:
  - Slot 1: 5°
  - Slot 30: 15°
  - Slot 23: 25°
  - Slot 16: 35°
  - Slot 9: 45°
  - Slot 2: 55°
  - Total: 6 slots
- **Phase C-** $[120°, 180°]$:
  - Slot 25: 125°
  - Slot 18: 135°
  - Slot 11: 145°
  - Slot 4: 155°
  - Slot 33: 165°
  - Slot 26: 175°
  - Total: 6 slots

Each sector has exactly 6 slots, confirming our calculations.
This is expected: $Q/(m \times t) = 36/(3 \times 1) = 12$ slots per phase total, split into 6 positive + 6 negative.

1.10.8 STEP 7: DETERMINE COIL THROW (y_q)

For double-layer winding: Theoretical pole pitch $= Q/(2p) = 36/10 = 3.6$ slots

We can choose:

- $y_q = 3$ (short pitch, ∼83% of pole pitch)
- $y_q = 4$ (near full pitch, ∼111% of pole pitch)

Let's choose $y_q = 4$ slots (commonly used for this config)

1.10.9 STEP 8: GENERATE COILS FOR EACH PHASE (DOUBLE LAYER)

Phase A:

- Positive sector slots: $[31, 24, 17, 10, 3, 32]$
  - Coil 1: begin = 31, end = 31 + 4 = 35, nc = +1
  - Coil 2: begin = 24, end = 24 + 4 = 28, nc = +1
  - Coil 3: begin = 17, end = 17 + 4 = 21, nc = +1

  - Coil 4: begin = 10, end = 10 + 4 = 14, nc = +1
  - Coil 5: begin = 3, end = 3 + 4 = 7, nc = +1
  - Coil 6: begin = 32, end = 32 + 4 = 36, nc = +1

- Negative sector slots: $[13, 6, 35, 28, 21, 14]$
  - Coil 7: begin = 13, end = 13 + 4 = 17, nc = -1
  - Coil 8: begin = 6, end = 6 + 4 = 10, nc = -1
  - Coil 9: begin = 35, end = 35 + 4 = 39 $\rightarrow$ 39 - 36 = 3, nc = -1
  - Coil 10: begin = 28, end = 28 + 4 = 32, nc = -1
  - Coil 11: begin = 21, end = 21 + 4 = 25, nc = -1
  - Coil 12: begin = 14, end = 14 + 4 = 18, nc = -1

  Phase A total: 12 coils

Phase B:

- Positive sector slots: $[19, 12, 5, 34, 27, 20]$
  - Coil 1: begin = 19, end = 19 + 4 = 23, nc = +1
  - Coil 2: begin = 12, end = 12 + 4 = 16, nc = +1
  - Coil 3: begin = 5, end = 5 + 4 = 9, nc = +1
  - Coil 4: begin = 34, end = 34 + 4 = 38 $\rightarrow$ 38 - 36 = 2, nc = +1
  - Coil 5: begin = 27, end = 27 + 4 = 31, nc = +1
  - Coil 6: begin = 20, end = 20 + 4 = 24, nc = +1

- Negative sector slots: $[1, 30, 23, 16, 9, 2]$
  - Coil 7: begin = 1, end = 1 + 4 = 5, nc = -1
  - Coil 8: begin = 30, end = 30 + 4 = 34, nc = -1
  - Coil 9: begin = 23, end = 23 + 4 = 27, nc = -1
  - Coil 10: begin = 16, end = 16 + 4 = 20, nc = -1
  - Coil 11: begin = 9, end = 9 + 4 = 13, nc = -1
  - Coil 12: begin = 2, end = 2 + 4 = 6, nc = -1

  Phase B total: 12 coils

Phase C:

- Positive sector slots: $[7, 36, 29, 22, 15, 8]$
  - Coil 1: begin = 7, end = 7 + 4 = 11, nc = +1
  - Coil 2: begin = 36, end = 36 + 4 = 40 $\rightarrow$ 40 - 36 = 4, nc = +1
  - Coil 3: begin = 29, end = 29 + 4 = 33, nc = +1
  - Coil 4: begin = 22, end = 22 + 4 = 26, nc = +1

  - Coil 5: begin = 15, end = 15 + 4 = 19, nc = +1
  - Coil 6: begin = 8, end = 8 + 4 = 12, nc = +1

- Negative sector slots: $[25, 18, 11, 4, 33, 26]$
  - Coil 7: begin = 25, end = 25 + 4 = 29, nc = -1
  - Coil 8: begin = 18, end = 18 + 4 = 22, nc = -1
  - Coil 9: begin = 11, end = 11 + 4 = 15, nc = -1
  - Coil 10: begin = 4, end = 4 + 4 = 8, nc = -1
  - Coil 11: begin = 33, end = 33 + 4 = 37 → 37 - 36 = 1, nc = -1
  - Coil 12: begin = 26, end = 26 + 4 = 30, nc = -1

  Phase C total: 12 coils

Total coils for all phases: 36 coils

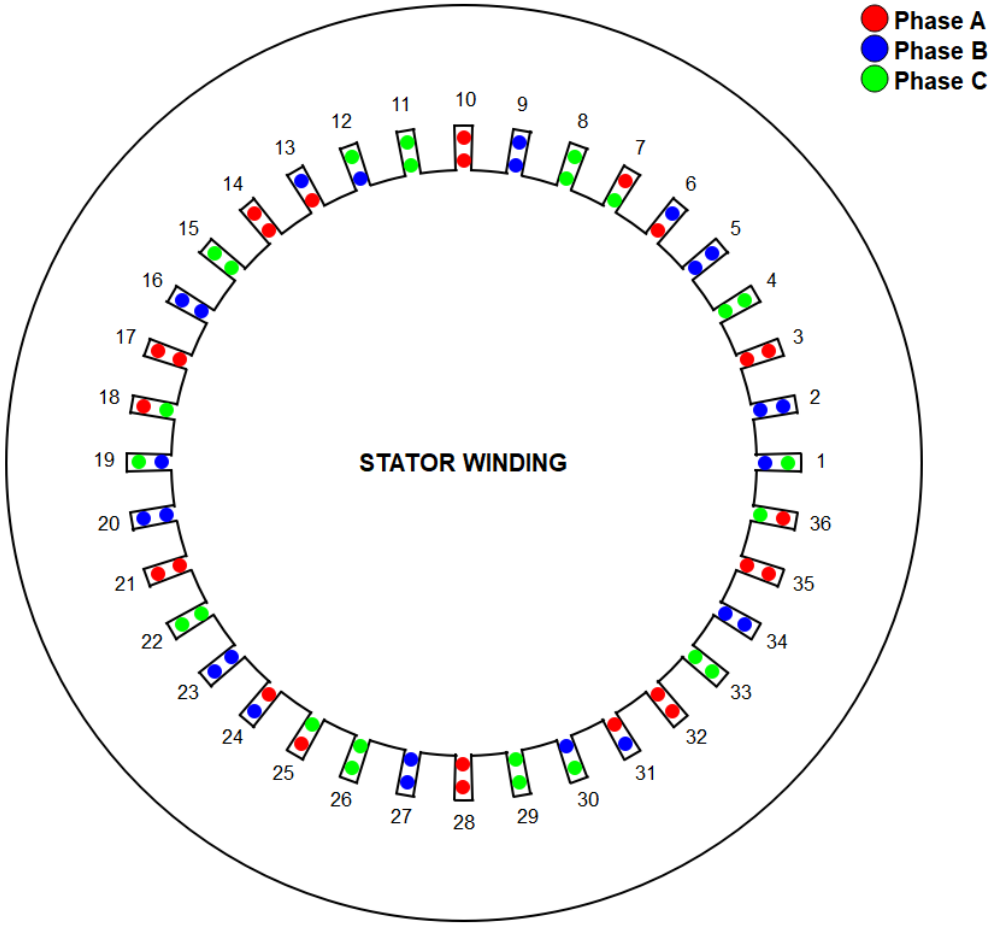


Figure 1.1: 36 slots,5 pair poles,3 phases, Double layer

1.10.10 STEP 9: CREATE DOUBLE LAYER MATRIX (6×36)

Matrix structure:

- Row 0: Phase A top layer
- Row 1: Phase B top layer
- Row 2: Phase C top layer
- Row 3: Phase A bottom layer
- Row 4: Phase B bottom layer
- Row 5: Phase C bottom layer

For each coil:

- Top layer (begin slot): Add $+nc$ to corresponding row and column
- Bottom layer (end slot): Add $-nc$ to corresponding row and column
  (Note: opposite sign because current direction reverses)

Let's build the matrix (showing non-zero entries):
Phase A (Row 0 = top, Row 3 = bottom):

- Coil(31→35, +1): Top[31]=+1, Bot[35]=-1
- Coil(24→28, +1): Top[24]=+1, Bot[28]=-1
- Coil(17→21, +1): Top[17]=+1, Bot[21]=-1
- Coil(10→14, +1): Top[10]=+1, Bot[14]=-1
- Coil(3→7, +1): Top[3]=+1, Bot[7]=-1
- Coil(32→36, +1): Top[32]=+1, Bot[36]=-1
- Coil(13→17, -1): Top[13]=-1, Bot[17]=+1
- Coil(6→10, -1): Top[6]=-1, Bot[10]=+1
- Coil(35→3, -1): Top[35]=-1, Bot[3]=+1
- Coil(28→32, -1): Top[28]=-1, Bot[32]=+1
- Coil(21→25, -1): Top[21]=-1, Bot[25]=+1
- Coil(14→18, -1): Top[14]=-1, Bot[18]=+1

Similarly for Phase B (Rows 1, 4) and Phase C (Rows 2, 5).
The resulting matrix shows the complete double-layer winding pattern:

- Positive values = current flowing INTO the page (or one direction)
- Negative values = current flowing OUT of the page (opposite direction)
- Each slot has TWO entries: one in top layer, one in bottom layer

1.10.11 STEP 10: GENERATE WINDING PATTERN FOR VISUALIZATION

For each slot, determine which phase occupies the top and bottom layers:

- Slot 1: Top = B+ (from Coil begin=1), Bottom = C- (from Coil end=33→1)
- Slot 2: Top = B- (from Coil begin=2), Bottom = B+ (from Coil end=34→2)
- Slot 3: Top = A+ (from Coil begin=3), Bottom = A- (from Coil end=35→3)
- ...and so on for all 36 slots.

The pattern alternates between phases in a specific sequence that creates the desired MMF distribution around the stator.

1.10.12 STEP 11: CALCULATE WINDING FACTORS

Given: $m = 3, Q = 36, p = 5, y_q = 4$

1. Slots per pole per phase:
$$q = \frac{36}{3 \times 10} = 1.2 = \frac{6}{5}$$

2. Electrical slot angle:
$$\alpha = \frac{2\pi \times p}{Q} = \frac{2\pi \times 5}{36} = \frac{10\pi}{36} = \frac{5\pi}{18} \text{ rad}$$
$\alpha = 50°$ electrical

3. Distribution factor (fundamental):
$$k_d = \frac{\sin(q \times \alpha/2)}{q \times \sin(\alpha/2)} = \frac{\sin(1.2 \times 50°/2)}{1.2 \times \sin(50°/2)} = \frac{\sin(30°)}{1.2 \times \sin(25°)} = \frac{0.5}{1.2 \times 0.4226} = \frac{0.5}{0.5071} = 0.9860$$

4. Pitch factor: Coil pitch in electrical radians: $\gamma = y_q \times \alpha$
$$\gamma = 4 \times \frac{5\pi}{18} = \frac{20\pi}{18} = \frac{10\pi}{9} \text{ rad}$$
$\gamma = 200°$ electrical
$$k_p = \sin(\gamma/2) = \sin(100°) = 0.9848$$

5. Skew factor (assuming no skew):
$$k_s = 1.0$$

6. Total winding factor:
$$k_w = k_d \times k_p \times k_s = 0.9860 \times 0.9848 \times 1.0 = 0.9709 \approx 0.971$$

This is an EXCELLENT winding factor (> 97% efficiency).

1.10.13 STEP 12: PERFORMANCE PREDICTIONS

1. Synchronous speed (for 50 Hz):

$$N = \frac{120 \times f}{P} = \frac{120 \times 50}{10} = 600 \text{ RPM}$$

2. Cogging torque: With fractional slot ($q = 6/5$), cogging is REDUCED compared to integer slot.
   Expected: LOW cogging torque

3. Torque ripple: With $Q/P = 3.6 > 3$, expected: LOW to MODERATE torque ripple

4. Harmonic content: Fractional slot reduces sub-harmonics and slot harmonics.
   Working harmonics: 1 (fundamental), 9, 11, 19, 21, ...
   (Harmonics of order: 1 ± 2, $1 \pm 2 \times (10/\gcd(Q, P))$, ...)

5. Winding unbalance: With $t = 1$ and perfect sector distribution: PERFECTLY BALANCED

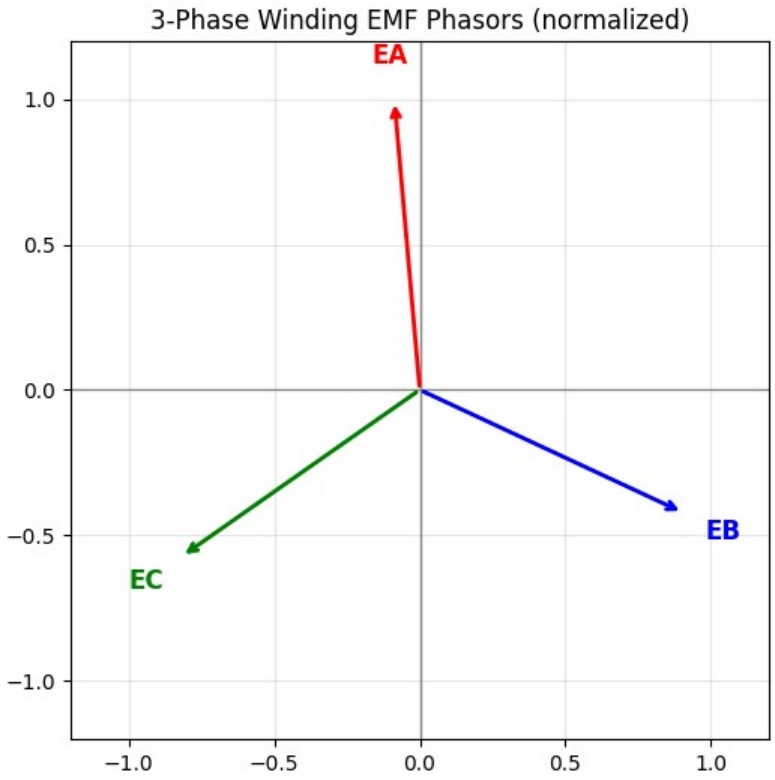


Figure 1.2: Voltage check

1.10.14 STEP 13: ALTERNATIVE - SINGLE LAYER WINDING

For comparison, let's briefly describe single-layer:

Coil throw must be ODD: $y_q = 3$ slots (since 4 is even)
Only ODD slots are used as starting points: Available slots: 1, 3, 5, 7, 9, ..., 35
Phase A positive: Use ODD slots from [31, 24, 17, 10, 3, 32]
→ Only: [31, 17, 3]
Phase A negative: Use ODD slots from [13, 6, 35, 28, 21, 14]
→ Only: [13, 35, 21]
Phase A coils (single layer):

- Coil(31→34, +1), Coil(17→20, +1), Coil(3→6, +1)
- Coil(13→16, -1), Coil(35→38≡2, -1), Coil(21→24, -1)

Total: 6 coils per phase × 3 phases = 18 coils (vs 36 for double layer)
Matrix: 3 × 36 (only one layer per phase)

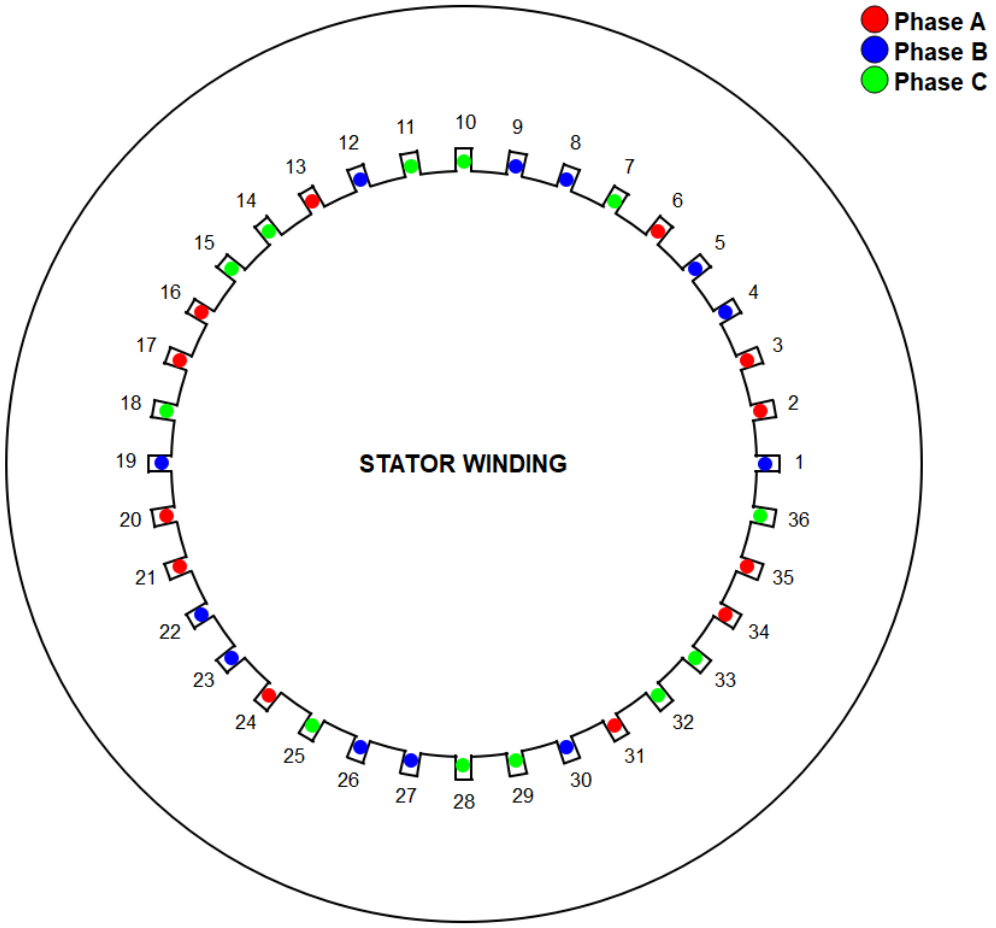


Figure 1.3: 36 slots,5 pair poles,3 phases, single layer

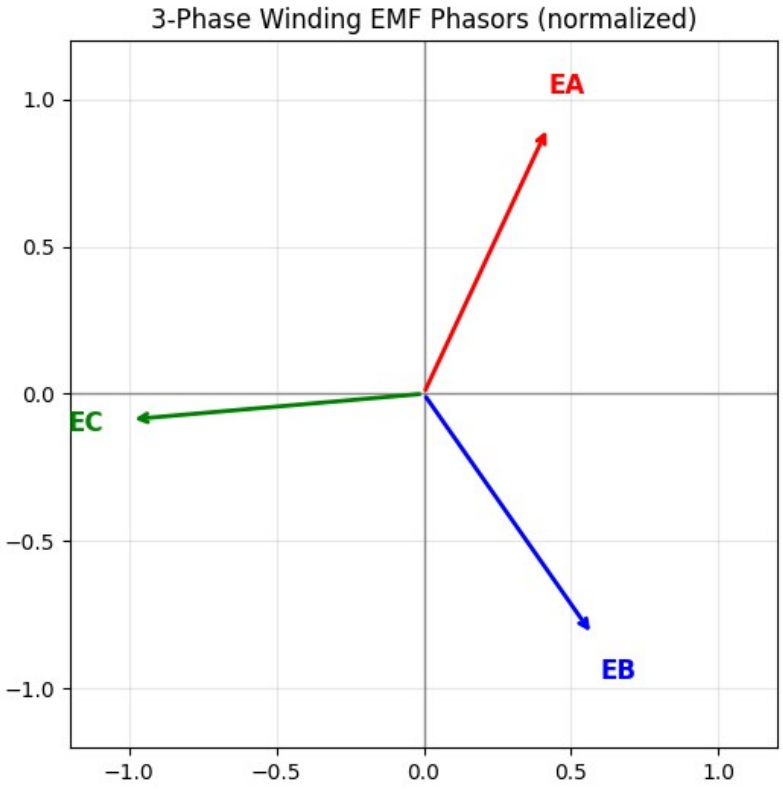


Figure 1.4: Voltage check

Single layer advantages:

- Simpler construction
- Better insulation between phases

Single layer disadvantages:

- Lower slot utilization (∼50%)
- Fixed coil span (must be odd)
- Less flexibility in design

1.11 CONCLUSION

This comprehensive explanation covered:

1. All fundamental parameters (q, t, τ_p, α, etc.)
2. Feasibility checks for balanced and single-layer windings
3. Star of slots theory with epsilon offset calculation
4. Sector generation and slot assignment

5. Coil generation for double and single layer
6. Winding factors (k_d, k_p, k_s, k_w)
7. Complete worked example for 3-phase, 36-slot, 5-pole-pair machine

The algorithm is UNIVERSAL and works for:

- Any number of phases ($m \geq 2$)
- Any number of slots (Q)
- Any number of pole pairs (p)
- Integer AND fractional slot windings
- Single AND double layer configurations

The key insight is using the STAR OF SLOTS in the ELECTRICAL coordinate system, with careful epsilon offsetting to avoid boundary ambiguities, and sector-based slot assignment to ensure perfect phase balance and symmetry.

This produces windings with high winding factors (typically $k_w > 0.90$), low harmonic content, reduced cogging torque, and balanced electromagnetic forces, resulting in smooth torque production and efficient machine operation.

Chapter 2

Fractional winding with $q < 1$

2.1 INTRODUCTION

When the slots per pole per phase (q) is less than 1, we enter the realm of FRACTIONAL SLOT CONCENTRATED WINDINGS (FSCW), also known as TOOTH-COIL WINDINGS or SUB-SLOT WINDINGS. These are fundamentally different from traditional distributed windings ($q \geq 1$) and have unique characteristics, advantages, and design considerations.

This chapter provides a comprehensive analysis of $q < 1$ windings, including:

- Mathematical foundations and parameter calculations
- Design methodology and constraints
- Worked examples with different q values
- Performance characteristics and trade-offs
- Practical applications and use cases

2.2 WHAT DOES $q < 1$ MEAN?

2.2.1 DEFINITION:

Recall that q (slots per pole per phase) is defined as:

$$q = \frac{Q}{m \times P} = \frac{Q}{m \times 2p}$$

where:

- Q = total number of slots
- m = number of phases
- P = number of poles = $2p$
- p = number of pole pairs

When $q < 1$, it means:

$$\frac{Q}{m \times 2p} < 1 \quad \text{Therefore: } Q < m \times 2p$$

In words: The number of slots is LESS than the number of phase belts

A phase belt is a region under one pole that should be occupied by one phase. With traditional distributed windings, each phase belt contains multiple slots ($q = 1, 2, 3, \dots$). With $q < 1$, we have MORE phase belts than available slots!

2.2.2 PHYSICAL INTERPRETATION:

Example: Consider a 3-phase machine with 12 slots and 14 poles (7 pole pairs)

$$q = \frac{12}{3 \times 14} = \frac{12}{42} = \frac{2}{7} \approx 0.286$$

This means:

- Total phase belts needed: 3 phases $\times$ 14 poles = 42 phase belts
- Available slots: 12 slots
- Each slot must "serve" multiple phase belts: $42/12 = 3.5$ phase belts per slot

Since we have more poles than slots (relative to phases), each coil spans **ONLY ONE TOOTH** (or very few teeth), hence the name "tooth-coil" winding.

Notice these are all PROPER FRACTIONS (numerator ¡ denominator).

2.3 CHARACTERISTICS OF $q < 1$ WINDINGS

2.3.1 CONCENTRATED COILS:

With $q < 1$, coils become CONCENTRATED rather than distributed:

- **Coil span is minimal** (typically 1 tooth = 1 slot pitch)
- Each coil wraps around a SINGLE TOOTH
- No overlapping of coils from the same phase
- Very SHORT END WINDINGS (major advantage)

Example: $y_q = 1$ slot (coil from slot N to slot $N + 1$)
This is drastically different from distributed windings where:

- $q \geq 1$: Coils span multiple teeth
- $y_q = 3 - 5$ slots typically
- Coils overlap within same phase
- LONG end windings

2.3.2 HIGH POLE COUNT:

Since $Q < m \times 2p$, for a given slot count, we must have:

$$p > \frac{Q}{2m}$$

For 3-phase machine with 12 slots:

$$p > \frac{12}{6} = 2$$

So minimum is $p = 3$ pole pairs (6 poles) for $q = 1$
To get $q < 1$, we need $p \geq 3$ (6+ poles)
Typical: $p = 4, 5, 6, 7, 8, \ldots$ (8, 10, 12, 14, 16, ... poles)
This leads to HIGH POLE COUNT designs, which:

- Reduce magnetic core size (smaller pole pitch)
- Increase electrical frequency ($f = p \times n/60$)
- Enable compact, high-torque-density machines
- Well-suited for direct-drive applications

2.3.3 FRACTIONAL SLOT COMBINATION:

The ratio Q/p determines the fundamental properties:

$$Q/p = \text{slot-to-pole ratio}$$

For $q < 1$ with m phases:

$$\frac{Q}{p} < 2m$$

Common ratios for 3-phase ($m = 3$):

- $Q/p = 6/7$ (e.g., $Q = 12, p = 14 \rightarrow q = 2/7$)
- $Q/p = 3/2$ (e.g., $Q = 9, p = 6 \rightarrow q = 1/2$)
- $Q/p = 6/5$ (e.g., $Q = 12, p = 10 \rightarrow q = 2/5$)
- $Q/p = 4/5$ (e.g., $Q = 12, p = 15 \rightarrow q = 4/15$)

The slot-pole combination creates a UNIQUE MMF HARMONIC SPECTRUM.

2.3.4 WINDING DISTRIBUTION:

With $q < 1$, the winding is NOT evenly distributed:

- Some teeth may have coils from one phase
- Adjacent teeth may have coils from different phases
- Pattern is highly asymmetric on a per-pole basis

- Symmetry exists only over multiple pole pairs (periodicity t)

Example: 12-slot, 10-pole, 3-phase ($q = 2/5$)

- Not every pole has all three phases represented equally
- Pattern repeats every $t = \gcd(12, 5) = 1$ periods (once around machine)

2.4 FUNDAMENTAL PARAMETER CALCULATIONS FOR $q < 1$

The same fundamental formulas apply, but with critical differences:

2.4.1 SLOTS PER POLE PER PHASE (q):

$$q = \frac{Q}{m \times 2p} < 1$$

Express as IRREDUCIBLE FRACTION:

$$q = \frac{n}{d} \quad \text{where } \gcd(n, d) = 1$$

Example: $Q = 12, m = 3, p = 5$

$$q = \frac{12}{3 \times 10} = \frac{12}{30} = \frac{2}{5}, \quad n = 2, d = 5$$

The denominator 'd' is particularly important - it indicates pattern complexity.

2.4.2 MACHINE PERIODICITY (t):

$$t = \gcd(Q, p)$$

This is CRITICAL for $q < 1$ windings because:

- Pattern may not repeat every pole pair
- Symmetry depends on t
- Unbalance can occur if t is not properly accounted for

Example: $Q = 12, p = 5$

$$t = \gcd(12, 5) = 1 \quad \text{(Pattern repeats once)}$$

Example: $Q = 12, p = 4$

$$t = \gcd(12, 4) = 4 \quad \text{(Pattern repeats 4 times around machine)}$$

2.4.3 COIL THROW (y_q):

For CONCENTRATED windings ($q < 1$):

$$y_q = 1 \text{ slot} \quad \text{(most common - single tooth coil)}$$

Or slightly distributed:

$$y_q = 2 \text{ or } 3 \text{ slots} \quad \text{(less common, semi-concentrated)}$$

The theoretical pole pitch is:

$$\tau_p = \frac{Q}{2p}$$

For $q < 1$: τ_p is typically 1-3 slots
Example: $Q = 12, p = 5$

$$\tau_p = \frac{12}{10} = 1.2 \text{ slots} \quad \text{Choose: } y_q = 1 \text{ slot (concentrated)}$$

2.4.4 ELECTRICAL SLOT ANGLE:

$$\alpha_s = \frac{360^\circ \times p}{Q} \quad \text{[degrees]}$$

$$\alpha_s = \frac{2\pi \times p}{Q} \quad \text{[radians]}$$

For $q < 1$, this angle is LARGE (many poles, few slots):
Example: $Q = 12, p = 5$

$$\alpha_s = \frac{360^\circ \times 5}{12} = 150^\circ \text{ electrical}$$

Example: $Q = 12, p = 7$

$$\alpha_s = \frac{360^\circ \times 7}{12} = 210^\circ \text{ electrical}$$

Compare to distributed winding ($Q = 36, p = 3$):

$$\alpha_s = \frac{360^\circ \times 3}{36} = 30^\circ \text{ electrical}$$

The large slot angle means:

- Adjacent slots have large electrical separation
- Less smooth MMF distribution
- Higher harmonic content

2.4.5 NUMBER OF COILS PER PHASE:

For double-layer winding:

$$\text{Coils per phase} = \frac{Q}{m}$$

Example: $Q = 12, m = 3$

$$\text{Coils per phase} = \frac{12}{3} = 4 \text{ coils}$$

For $q < 1$, this is SMALL compared to distributed windings.

2.5 BALANCE AND FEASIBILITY FOR $q < 1$

2.5.1 BALANCE CONDITION:

The same balance condition applies:

$$\frac{Q}{m \times t} = \text{INTEGER}$$

where $t = \gcd(Q, p)$

This ensures each phase gets equal number of slots per repeating period.
For $q < 1$, this is MORE RESTRICTIVE because:

- Small Q makes it harder to divide evenly
- High p creates small t values
- Easy to create unbalanced windings

2.5.2 CHECKING BALANCE:

Example 1: $Q = 12, m = 3, p = 5$

$$t = \gcd(12, 5) = 1, \quad \frac{Q}{m \times t} = \frac{12}{3 \times 1} = 4 \quad \text{INTEGER} \rightarrow \text{BALANCED}$$

Example 2: $Q = 12, m = 3, p = 7$

$$t = \gcd(12, 7) = 1, \quad \frac{Q}{m \times t} = \frac{12}{3 \times 1} = 4 \quad \text{INTEGER} \rightarrow \text{BALANCED}$$

Example 3: $Q = 10, m = 3, p = 4$

$$t = \gcd(10, 4) = 2, \quad \frac{Q}{m \times t} = \frac{10}{3 \times 2} = 1.667\ldots \quad \text{NOT INTEGER} \rightarrow \text{NOT BALANCED}$$

2.5.3 MINIMUM q CONSTRAINT:

While mathematically q can be very small, practically:

$$q \geq 0.25 \quad \text{(preferred)}, \quad q \geq 0.20 \quad \text{(absolute minimum)}$$

Below this, the winding becomes:

- Too concentrated (excessive harmonic content)
- Poor fundamental winding factor
- High torque ripple
- Thermal issues (hot spots on few teeth)

Example: $Q = 12, m = 3, p = 10$ (20 poles)

$$q = \frac{12}{60} = 0.2 \quad \rightarrow \text{At the limit}$$

Example: $Q = 12, m = 3, p = 12$ (24 poles)

$$q = \frac{12}{72} = 0.167 \quad \rightarrow \text{TOO LOW, not recommended}$$

2.6 WINDING PATTERN GENERATION FOR $q < 1$

The STAR OF SLOTS algorithm works identically for $q < 1$, but with notable differences in behavior:

STEP 1: CREATE STAR OF SLOTS

Each slot i has electrical angle:

$$\theta_i = \alpha_s \times (i - 1) + \varepsilon$$

With large α_s (due to high pole count), the spokes are MORE WIDELY SPACED in electrical degrees, even though mechanically they're evenly distributed.

2.6.1 STEP 2: EPSILON CALCULATION

The epsilon offset calculation uses the same rules, but:

$$\text{spoke_per_phase} = \frac{Q}{m \times t}$$

For $q < 1$, this value is typically SMALL (1, 2, 3, or 4).
Example: $Q = 12, m = 3, p = 5, t = 1$

$$\text{spoke_per_phase} = \frac{12}{3 \times 1} = 4$$

Since $m = 3$ (odd) and spoke_per_phase= 4 (even):

$$\varepsilon = \frac{\alpha_{ph}}{2} = \frac{(2\pi/Q \times t)}{2} = \frac{\pi}{Q} \times t$$

$$\varepsilon = \frac{\pi}{12} \times 1 = 15^\circ \text{ electrical}$$

2.6.2 STEP 3: SECTOR GENERATION

Sectors are created identically:

$$\text{Each sector spans: } \frac{180^\circ}{m}$$

For 3-phase: 60° sectors
However, with LARGE α_s and FEW slots, many sectors contain:

- Only 1 slot
- Sometimes 0 slots (if slot lands in different sector)
- Rarely 2+ slots

This creates the concentrated, non-distributed nature.

2.6.3 STEP 4: SLOT ASSIGNMENT

With large electrical spacing between slots, assignment becomes:

- More deterministic (less ambiguity)
- Highly dependent on epsilon (small errors matter more)
- Creates distinct phase zones

Example: $Q = 12, p = 5, \alpha_s = 150^\circ$
Slot angles (with $\varepsilon = 15^\circ$):

- Slot 1: 15°
- Slot 2: 165°
- Slot 3: $315^\circ \equiv -45^\circ + 360^\circ$
- Slot 4: $465^\circ \equiv 105^\circ$
- Slot 5: $615^\circ \equiv 255^\circ$
- Slot 6: $765^\circ \equiv 45^\circ$
- Slot 7: $915^\circ \equiv 195^\circ$
- Slot 8: $1065^\circ \equiv 345^\circ$
- Slot 9: $1215^\circ \equiv 135^\circ$
- Slot 10: $1365^\circ \equiv 285^\circ$
- Slot 11: $1515^\circ \equiv 75^\circ$
- Slot 12: $1665^\circ \equiv 225^\circ$

Phase A+ sector $[60^\circ, 120^\circ]$:

$$\text{Slots: } 4(105^\circ), 11(75^\circ) \rightarrow 2 \text{ slots}$$

Phase A- sector $[240°, 300°]$:

$$\text{Slots: } 5(255°), 10(285°) \rightarrow 2 \text{ slots}$$

Phase B+ sector $[180°, 240°]$:

$$\text{Slots: } 7(195°), 12(225°) \rightarrow 2 \text{ slots}$$

Phase B- sector $[0°, 60°]$:

$$\text{Slots: } 1(15°), 6(45°) \rightarrow 2 \text{ slots}$$

Phase C+ sector $[300°, 360°]$:

$$\text{Slots: } 3(315°), 8(345°) \rightarrow 2 \text{ slots}$$

Phase C- sector $[120°, 180°]$:

$$\text{Slots: } 9(135°), 2(165°) \rightarrow 2 \text{ slots}$$

Perfect balance: Each sector has 2 slots.
Total: $2 + 2 = 4$ slots per phase (confirms $Q/m = 12/3 = 4$)

2.6.4 STEP 5: COIL GENERATION

For concentrated winding with $y_q = 1$:
Phase A coils:

- Positive: Coil(4→5, +1), Coil(11→12, +1)
- Negative: Coil(5→6, -1), Coil(10→11, -1)

Phase B coils:

- Positive: Coil(7→8, +1), Coil(12→1, +1)
- Negative: Coil(1→2, -1), Coil(6→7, -1)

Phase C coils:

- Positive: Coil(3→4, +1), Coil(8→9, +1)
- Negative: Coil(9→10, -1), Coil(2→3, -1)

Total: 12 coils (one per slot, as expected for concentrated winding)
Notice:

- Each coil spans exactly 1 slot (tooth-coil)
- No overlapping coils
- Every slot has one coil beginning in it
- Coils alternate between phases around the circumference

WINDING FACTORS FOR $q < 1$ WINDINGS

Winding factors for $q < 1$ are FUNDAMENTALLY DIFFERENT from distributed windings:

2.6.5 DISTRIBUTION FACTOR (k_d):

The proper formula for fractional $q < 1$:

$$k_d = \frac{|\sin(\pi \times p/Q \times Q/(m \times p))|}{Q/(m \times 2p) \times m \times |\sin(\pi \times p/Q)|}$$

This simplifies based on specific geometry.
PRACTICAL VALUES for $q < 1$:

- $q = 0.5$: $k_d \approx 0.95 - 0.97$
- $q = 0.4$: $k_d \approx 0.93 - 0.95$
- $q = 0.33$: $k_d \approx 0.90 - 0.93$
- $q = 0.25$: $k_d \approx 0.85 - 0.90$

Lower $q \rightarrow$ Lower k_d (more concentration $\rightarrow$ less effective distribution)

2. PITCH FACTOR (k_p):

For concentrated winding with $y_q = 1$:

$$\text{Coil pitch: } \gamma = y_q \times \alpha = 1 \times \alpha$$

Example: $\alpha = 150^\circ$

$$\gamma = 150^\circ \text{ electrical}, \quad k_p = \sin(\gamma/2) = \sin(75^\circ) = 0.966$$

For comparison, full-pitch: $\gamma = 180^\circ$, $k_p = \sin(90^\circ) = 1.0$
So concentrated windings are inherently SHORT-PITCHED, which:

- Reduces fundamental k_p (bad for torque)
- Helps eliminate certain harmonics (good for quality)

2.6.6 TOTAL WINDING FACTOR (k_w):

$$k_w = k_d \times k_p \times k_s$$

For typical $q < 1$ concentrated winding:

$$k_d \approx 0.90 - 0.95, \quad k_p \approx 0.87 - 0.97, \quad k_s \approx 1.0 \text{ (usually no skew)}$$

Result: $k_w \approx 0.78 - 0.92$
This is LOWER than good distributed windings ($k_w \approx 0.90 - 0.96$).

Example: $q = 2/5, y_q = 1, \alpha = 150°$

$$k_d \approx 0.933, \quad k_p = 0.966, \quad k_w = 0.933 \times 0.966 = 0.901$$

Example: $q = 1/2, y_q = 1, \alpha = 120°$

$$k_d \approx 0.955, \quad k_p = 0.866, \quad k_w = 0.955 \times 0.866 = 0.827$$

TRADE-OFF:
Lower winding factor means less torque per ampere (bad), BUT:

- Shorter end windings = less copper = less resistance (good)
- Higher pole count = smaller airgap flux density needed (good)
- Net result can be BETTER efficiency despite lower k_w

2.7 HARMONIC CONTENT AND MMF SPECTRUM FOR $q < 1$

The MMF harmonic spectrum of $q < 1$ windings is RADICALLY DIFFERENT from distributed windings. This is both an advantage AND a challenge.

WORKING HARMONICS:

The main working harmonics are determined by:

$$\nu = 1 \pm \frac{2km}{\gcd(Q, p)}$$

where $k = 0, 1, 2, 3, \ldots$

Example: $Q = 12, p = 5, m = 3, \gcd(12, 5) = 1$

$$\nu = 1 \pm 6k$$

Working harmonics: $\nu = 1, 7, 13, 19, 25, \ldots$
(Also: $5, 11, 17, 23, \ldots$ from negative)

Simplified: Main harmonics are $1, 5, 7, 11, 13, 17, 19, \ldots$

2.7.1 LOW-ORDER HARMONICS:

A critical issue with $q < 1$ is SUB-HARMONICS and low-order space harmonics:

Sub-harmonics: Harmonics with order $\nu < p$ (pole pairs)

Example: $p = 5$, harmonics: 1 (sub), 5 (principal), 7, 11, 13, ...
The $\nu = 1$ space harmonic has only 1 pole pair!
This can cause unwanted forces, vibrations, and losses.

For $q < 1$:

- Sub-harmonics are STRONG
- Low-order harmonics (5th, 7th, 11th) are significant
- Can cause torque ripple and acoustic noise

2.7.2 HARMONIC CANCELLATION:

Unlike distributed windings where harmonics are "distributed away," in $q < 1$:

- Harmonics must be managed by slot-pole combination
- Some combinations naturally suppress certain harmonics
- Careful selection of Q and p is CRITICAL

Good combinations: $\gcd(Q, p) = 1$ and Q/p close to special ratios
Bad combinations: $\gcd(Q, p) > 1$ (creates strong sub-harmonics)

2.7.3 ROTOR LOSSES:

High harmonic content causes:

- Eddy current losses in rotor (magnets or bars)
- Heating and demagnetization risk
- Reduced efficiency

Mitigation:

- Segmented magnets (for PM machines)
- Rotor skewing
- Optimized slot-pole combination
- Laminated rotor (for induction machines)

2.8 ADVANTAGES AND DISADVANTAGES OF $q < 1$ WINDINGS

ADVANTAGES:

1. **SHORT END WINDINGS**
   - Minimal copper usage (20-40% less than distributed)
   - Reduced DC resistance $\rightarrow$ Lower I^2R losses
   - Better power density (more copper in active region)
   - Lower manufacturing cost
2. **HIGH FAULT TOLERANCE**
   - Phases are magnetically decoupled (no mutual inductance)
   - Single-phase fault doesn't affect other phases
   - Can continue operation with one phase open
   - Critical for safety applications (aerospace, EVs)

3. **MODULAR CONSTRUCTION**
    - Each tooth can be wound independently
    - Allows segmented stator design
    - Easier automated winding
    - Better for high-volume manufacturing
4. **COMPACT DESIGN**
    - Shorter axial length (short end windings)
    - Enables pancake/axial flux topologies
    - Better for space-constrained applications
    - Higher torque density potential
5. **FLEXIBLE POLE CONFIGURATION**
    - Easy to change pole count (just rearrange coils)
    - Can optimize for different speed ranges
    - Enables variable pole topologies
6. **LOWER COGGING (sometimes)**
    - Certain Q/p combinations have very low cogging
    - Fractional slot effect reduces cogging torque
    - Better than integer slot distributed windings

2.8.1 DISADVANTAGES:

1. **LOWER WINDING FACTOR**
    - $k_w = 0.75 - 0.90$ (vs $0.90 - 0.96$ for distributed)
    - Less torque per ampere
    - Requires more current for same torque
    - Larger inverter rating needed
2. **HIGH HARMONIC CONTENT**
    - Strong sub-harmonics ($\nu < p$)
    - Significant 5th, 7th, 11th harmonics
    - Causes torque ripple (5-15% typical)
    - Acoustic noise and vibration
3. **ROTOR LOSSES**
    - High-frequency harmonics induce eddy currents
    - 2-5× higher rotor losses than distributed

- Magnet heating and demagnetization risk
- Reduced efficiency at high speed

4. **RADIAL FORCES**
   - Unbalanced magnetic pull (UMP) risk
   - Asymmetric force distribution
   - Bearing stress and acoustic noise
   - Vibration issues
5. **SATURATION ISSUES**
   - Flux concentrated in fewer teeth
   - Tooth tip saturation limits torque
   - Requires larger teeth or lower flux density
   - Increases machine size
6. **LIMITED OPTIMIZATION**
   - Fewer discrete Q/p combinations available
   - Must satisfy balance conditions
   - Less flexibility than distributed windings
   - Harder to meet specific performance targets

2.9 COMPLETE WORKED EXAMPLE : $q = 2/5$

2.9.1 CONFIGURATION:

- $Q = 12$ slots
- $p = 5$ pole pairs (10 poles)
- $m = 3$ phases
- $y_q = 1$ slot (concentrated)

2.9.2 STEP 1: CALCULATE PARAMETERS

$$q = \frac{12}{3 \times 10} = \frac{12}{30} = \frac{2}{5} = 0.4$$

$$t = \gcd(12, 5) = 1 \quad \text{(no internal periodicity)}$$

$$\tau_p = \frac{12}{10} = 1.2 \text{ slots}$$

$$\alpha = \frac{360^\circ \times 5}{12} = 150^\circ$$

$$\alpha_{ph} = \frac{360^\circ \times 1}{12} = 30^\circ$$

2.9.3 STEP 2: CHECK FEASIBILITY

$$\text{Balance: } \frac{Q}{m \times t} = \frac{12}{3 \times 1} = 4 \quad \text{INTEGER} \rightarrow \text{BALANCED}$$

$$\text{Minimum } q : q = 0.4 \geq 0.25 \quad \text{ACCEPTABLE}$$

STEP 3: EPSILON OFFSET

$$\text{spoke_per_phase} = \frac{12}{3 \times 1} = 4$$

Since $m = 3$ (odd) and spoke_per_phase= 4 (even):

$$\varepsilon = \frac{\alpha_{ph}}{2} = \frac{30^\circ}{2} = 15^\circ$$

2.9.4 STEP 4: STAR OF SLOTS

Slot angles ($\alpha = 150^\circ, \varepsilon = 15^\circ$):

- Slot 1: 15°
- Slot 2: 165°
- Slot 3: 315°
- Slot 4: $465^\circ = 105^\circ$
- Slot 5: $615^\circ = 255^\circ$
- Slot 6: $765^\circ = 45^\circ$
- Slot 7: $915^\circ = 195^\circ$
- Slot 8: $1065^\circ = 345^\circ$
- Slot 9: $1215^\circ = 135^\circ$
- Slot 10: $1365^\circ = 285^\circ$
- Slot 11: $1515^\circ = 75^\circ$
- Slot 12: $1665^\circ = 225^\circ$

Sorted:

$$15^\circ(1), 45^\circ(6), 75^\circ(11), 105^\circ(4), 135^\circ(9), 165^\circ(2),$$
$$195^\circ(7), 225^\circ(12), 255^\circ(5), 285^\circ(10), 315^\circ(3), 345^\circ(8)$$

2.9.5 STEP 5: SECTOR ASSIGNMENT

- Phase A+ $[60^\circ, 120^\circ]$: Slots $\{11(75^\circ), 4(105^\circ)\}$
- Phase A- $[240^\circ, 300^\circ]$: Slots $\{5(255^\circ), 10(285^\circ)\}$
- Phase B+ $[180^\circ, 240^\circ]$: Slots $\{7(195^\circ), 12(225^\circ)\}$
- Phase B- $[0^\circ, 60^\circ]$: Slots $\{1(15^\circ), 6(45^\circ)\}$
- Phase C+ $[300^\circ, 360^\circ]$: Slots $\{3(315^\circ), 8(345^\circ)\}$
- Phase C- $[120^\circ, 180^\circ]$: Slots $\{9(135^\circ), 2(165^\circ)\}$

Perfect balance: Each phase has $2 + 2 = 4$ slots.

2.9.6 STEP 6: GENERATE COILS ($y_q = 1$)

- Phase A (4 coils): Positive: $\{11 \to 12, 4 \to 5\}$; Negative: $\{5 \to 6, 10 \to 11\}$
- Phase B (4 coils): Positive: $\{7 \to 8, 12 \to 1\}$; Negative: $\{1 \to 2, 6 \to 7\}$
- Phase C (4 coils): Positive: $\{3 \to 4, 8 \to 9\}$; Negative: $\{9 \to 10, 2 \to 3\}$

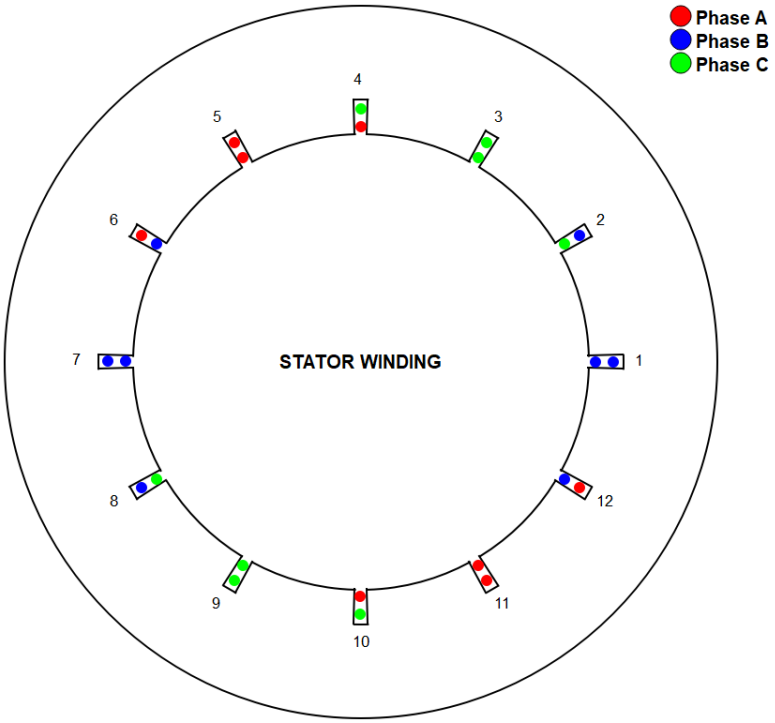


Figure 2.1: 12 slots, 5 pair poles, 3 phases, Double layer

2.9.7 STEP 7: WINDING FACTORS

$$k_d \text{ for } q = 0.4 : k_d \approx 0.933$$

$$k_p \text{ for } \alpha = 150^\circ : k_p = \sin(75^\circ) = 0.966$$

$$k_w = 0.933 \times 0.966 = 0.901$$

This is a good winding factor for $q < 1$!

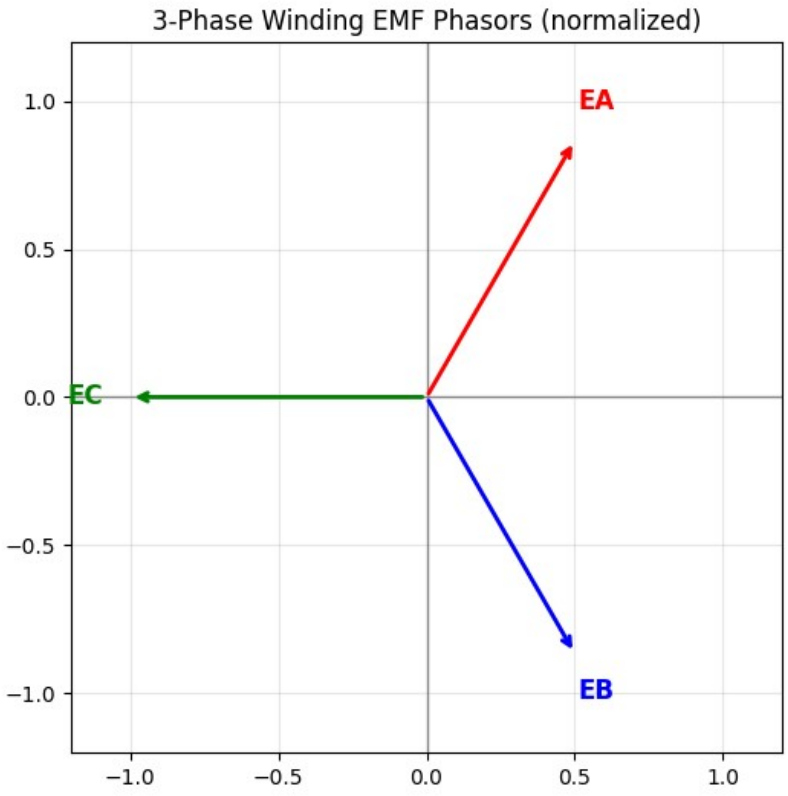


Figure 2.2: Voltage check

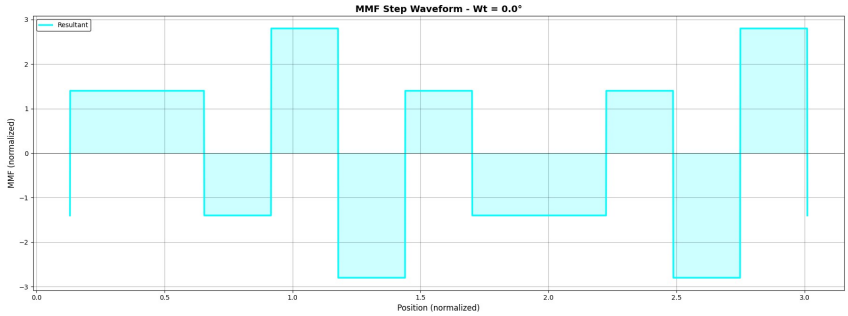


Figure 2.3: MMF for $q = 1/2$

2.9.8 STEP 8: WINDING PATTERN

Slot-by-slot pattern (Top layer only, since concentrated):

- Slot 1: B-
- Slot 2: B-
- Slot 3: C+
- Slot 4: C+
- Slot 5: A-
- Slot 6: A-
- Slot 7: B+
- Slot 8: B+
- Slot 9: C-
- Slot 10: C-
- Slot 11: A+
- Slot 12: A+

Notice the pattern: Each phase occupies 2 consecutive slots, then switches. This creates a concentrated, tooth-coil winding with clean phase separation.

2.10 PRACTICAL APPLICATIONS OF $q < 1$ WINDINGS

1. **DIRECT-DRIVE WIND TURBINES**
   - Large diameter, low speed
   - High pole count ($p = 40 - 100$)
   - $Q = 90 - 180$ slots
   - $q = 0.3 - 0.5$ typical
   - Benefits: Short end windings, modular construction
2. **IN-WHEEL ELECTRIC VEHICLE MOTORS**
   - Compact, pancake design
   - High torque density required
   - Typical: $Q = 12, p = 10 - 14$ ($q = 0.2 - 0.4$)
   - Benefits: Short axial length, fault tolerance
3. **AEROSPACE ACTUATORS**
   - High reliability critical
   - Fault tolerance essential
   - $Q = 12 - 18, p = 5 - 8$
   - Benefits: Phase decoupling, continue operation after fault
4. **SERVO DRIVES AND ROBOTICS**
   - High dynamic response needed
   - Low rotor inertia preferred
   - $q = 0.3 - 0.5$ common
   - Benefits: Low cogging, smooth torque (with proper design)
5. **AXIAL FLUX MACHINES**
   - Pancake geometry
   - Very short end windings critical
   - Concentrated windings ideal
   - Benefits: Maximize active material, minimize inactive
6. **LOW-SPEED HIGH-TORQUE DRIVES**
   - Ship propulsion, industrial drives
   - Direct drive (no gearbox)
   - High pole count ($p = 20 - 50$)
   - Benefits: High torque density, modular stator

2.11 DESIGN GUIDELINES FOR $q < 1$

2.11.1 SLOT-POLE COMBINATION SELECTION:

1. Choose Q to be divisible by m (number of phases)

$$Q = m \times k \quad \text{where } k \text{ is integer}$$

2. Select p such that $\gcd(Q, p) = 1$ or small value
   Smaller gcd $\rightarrow$ better harmonic performance
3. Ensure $q \geq 0.25$ (preferably $q \geq 0.33$)
4. Check balance: $Q/(m \times \gcd(Q, p)) = \text{integer}$
5. Preferred combinations for 3-phase:
   - 12-slot: 10-pole ($q = 2/5$), 8-pole ($q = 1/2$), 14-pole ($q = 2/7$)
   - 9-slot: 8-pole ($q = 3/8$), 10-pole ($q = 3/10$)
   - 18-slot: 16-pole ($q = 9/16$), 20-pole ($q = 3/10$)

2.11.2 MINIMIZE TORQUE RIPPLE:

- Choose Q/p ratio close to: $6/5, 3/2, 4/3$
- Avoid: $Q/p = 1$ or 2 (very high ripple)
- Use skewing (1 slot pitch) if needed
- Optimize slot opening width

2.11.3 REDUCE ROTOR LOSSES:

- Segmented magnets (axially and circumferentially)
- Lower airgap flux density
- Increase airgap length slightly
- Use laminated rotor (if applicable)

2.11.4 MANAGE SATURATION:

- Wider teeth (accept lower slot fill)
- Higher grade magnetic steel
- Lower flux density operation
- Consider dual-layer magnets (if PM)

2.11.5 THERMAL MANAGEMENT:

- Concentrated windings create hot spots
- Need good cooling for active teeth
- Consider liquid cooling for high power
- Thermal potting around coils

2.12 CONCLUSIONS

Fractional slot concentrated windings ($q < 1$) represent a distinctly different design paradigm from traditional distributed windings:

KEY CHARACTERISTICS:

- Tooth-coil construction with $y_q = 1 - 2$ slots
- High pole count relative to slot count
- Short end windings (20-40% copper savings)
- Lower winding factors ($k_w = 0.75 - 0.90$)
- High harmonic content and torque ripple
- Excellent fault tolerance

BEST SUITED FOR:

- Applications requiring compact axial length
- Direct-drive systems with many poles
- Safety-critical systems needing fault tolerance
- Modular manufacturing with automation
- Space-constrained applications

NOT SUITED FOR:

- Applications demanding very smooth torque
- High-speed operation (excessive rotor losses)
- When maximum torque density is critical
- Ultra-low noise applications

The star of slots algorithm handles $q < 1$ identically to $q \geq 1$, with the concentrated nature emerging naturally from the sector assignment when there are fewer slots than phase belts.

Successful $q < 1$ design requires careful slot-pole combination selection, attention to harmonic content, and appropriate thermal and mechanical design to handle the concentrated flux and forces.

Chapter 3

MMF Calculation Method - Application Overview

This chapter explains the method used to calculate the Magnetomotive Force (MMF) distribution in electric machine stator windings using a tooth-centered approach. The method is phase-independent and works for any number of phases.

3.1 Core Methodology

3.1.1 Slot Current Calculation

Winding Matrix Representation

- The winding matrix defines how coils are connected in slots
- Single layer: Matrix shape = $(m \times Q)$ where m = number of phases, Q = number of slots
- Double layer: Matrix shape = $(2m \times Q)$ with separate rows for top and bottom layers
- Matrix values: $+1$ (current going in), -1 (current coming out), 0 (no conductor)

Instantaneous Current Calculation

- Each phase has a time-varying current based on electrical angle W_t
- For phase k: $I_k(t) = I_{\text{amplitude}} \times \cos(W_t - 2\pi k/m)$
- Example for 3-phase at $W_t = 90°$:

$$\text{Phase A: } I_A = I \times \cos(90° - 0°) = 0$$
$$\text{Phase B: } I_B = I \times \cos(90° - 120°) = I \times \cos(-30°) = 0.866 \times I$$
$$\text{Phase C: } I_C = I \times \cos(90° - 240°) = I \times \cos(-150°) = -0.866 \times I$$

Slot MMF Contribution

- For each slot and phase: $\text{MMF}_{\text{slot}} = \text{Matrix_value} \times I_{\text{phase}} \times (N/2)$
- The factor $N/2$ accounts for the coil turns contributing to air gap MMF
- For double layer: Top and bottom contributions are summed per phase
- Total slot current = Sum of all phase contributions in that slot

3.1.2 Tooth MMF Calculation (Iterative Method)

Tooth-Centered Approach Rationale

- MMF varies around the air gap due to discrete slot positions
- Teeth are the magnetic flux paths; slots contain conductors
- Calculating at tooth centers gives accurate flux path representation

Initial Tooth MMF

- Find first non-zero slot current (starting reference point)
- Count consecutive slots with same current value around this point
- Initial tooth MMF formula:

$$\text{MMF}_{\text{tooth}}[\text{first}] = -\text{number}_{\text{same}} \times \text{slot}_{\text{current}}[\text{first}] + 2 \times \text{slot}_{\text{current}}[\text{first}] \times (\text{left}_{\text{count}} + 1)$$

- This accounts for the ampere-turns on both sides of the tooth

Iterative Calculation

- For subsequent teeth, apply Ampère's circuital law:

$$\text{MMF}_{\text{tooth}}[i+1] = \text{MMF}_{\text{tooth}}[i] + 2 \times \text{slot}_{\text{current}}[i+1]$$

- The factor of 2 appears because the slot current is crossed twice in the closed path
- This method ensures the MMF differences equal the enclosed current

Circular Array Handling

- The stator is circular, so index wrapping is applied
- Last slot connects back to first slot seamlessly
- Ensures continuity in the 360° profile

3.1.3 Continuous Profile Generation

Tooth Region (Constant MMF)

- Each tooth occupies an angular width = $\text{Tooth}_{\text{arc}}$
- $\text{Tooth}_{\text{arc}} = (360°/Q) - \text{Slot}_{\text{arc}}$
- Throughout the tooth width, MMF remains constant at tooth center value
- This represents the equipotential nature of the iron tooth

Slot Region (Linear Interpolation)

- Between two adjacent teeth, MMF changes linearly across the slot
- Interpolation parameter: $t = (\text{angle} - \text{tooth}_{\text{end}})/\text{slot}_{\text{width}}$, where $0 \leq t \leq 1$
- $\text{MMF}_{\text{slot}}(\text{angle}) = (1-t) \times \text{MMF}_{\text{tooth}}[i] + t \times \text{MMF}_{\text{tooth}}[i+1]$
- Represents the gradual change in magnetic potential across air gap

High-Resolution Sampling

- Default: 3600 angular points (0.1° resolution)
- Provides smooth curves for visualization
- Each point is classified as either in tooth region or slot region

3.1.4 Resultant MMF Calculation

Time-Domain Superposition

- Each phase creates its own MMF distribution
- At any instant (W_t angle), phases carry different instantaneous currents
- The magnetic fields superimpose linearly

Weighting by Sinusoidal Factors

- For m phases, phase k has weight: $\sin(W_t - 2\pi k/m)$
- General formula: $\text{MMF}_{\text{resultant}} = \sum_{k=0}^{m-1}[\text{MMF}_{\text{phase}_k} \times \sin(W_t - 2\pi k/m)]$

Three-Phase Example ($m = 3$)

- $\text{MMF}_{\text{resultant}} = \text{MMF}_A \times \sin(W_t) + \text{MMF}_B \times \sin(W_t - 120°) + \text{MMF}_C \times \sin(W_t - 240°)$
- Simplifies to: $\text{MMF}_A \times \sin(W_t) + \text{MMF}_B \times \sin(W_t + 120°) + \text{MMF}_C \times \sin(W_t - 120°)$
- Creates a rotating magnetic field

Multi-Phase Flexibility

- Works for any number of phases (3, 5, 6, 7, etc.)
- Phase separation: $360°/m$ electrical degrees
- Automatically adapts based on winding matrix dimensions

3.1.5 Harmonic Analysis (Discrete Fourier Transform)

Sampling at Tooth Centers

- Extract MMF values at Q tooth positions (one per slot pitch)
- These Q samples represent one complete mechanical revolution
- Tooth positions: $\theta_j = j \times (360°/Q)$ for $j = 0, 1, \ldots, Q-1$

DFT Coefficient Calculation

- For each harmonic order ν ($\nu = 1, 2, 3, \ldots, \nu_{\max}$):

$$a_\nu = \frac{2}{Q} \times \sum_{j=0}^{Q-1} [\text{MMF}_{\text{tooth}}[j] \times \cos(\nu \times \theta_j)]$$

$$b_\nu = \frac{2}{Q} \times \sum_{j=0}^{Q-1} [\text{MMF}_{\text{tooth}}[j] \times \sin(\nu \times \theta_j)]$$

- Normalization factor $2/Q$ gives correct amplitude scaling

Harmonic Amplitude and Phase

- Amplitude: $A_\nu = \sqrt{a_\nu^2 + b_\nu^2}$
- Phase angle: $\phi_\nu = \text{atan2}(b_\nu, a_\nu)$
- MMF reconstruction: $\text{MMF}(\theta) = \sum_\nu [A_\nu \times \cos(\nu \times \theta - \phi_\nu)]$

Percentage of Fundamental

- Fundamental amplitude: A_1 (first harmonic)
- Harmonic percentage: $(A_\nu / A_1) \times 100\%$
- Indicates winding quality (lower harmonics = better)

3.2 Key Parameters

- N: Number of turns per coil (e.g., 204 turns)
- I: Current amplitude in Amperes (e.g., 2.8 A)
- W_t: Electrical angle in degrees (0° to 360°), represents time position in AC cycle
- Q: Number of slots (e.g., 12, 24, 36 slots)
- Slot_{arc}: Angular width of each slot = $360°/(2 \times Q)$ degrees
- $\text{Tooth}_{\text{arc}}$: Angular width of each tooth = $(360°/Q) - \text{Slot}_{\text{arc}}$
- m: Number of phases (e.g., 3 for three-phase)
- Matrix: Winding connection pattern ($m \times Q$ or $2m \times Q$ array)

3.3 Complete Worked Example: 3-Phase, 12-Slot, Double-Layer Winding (1 Pole Pair)

3.3.1 Given Parameters

- Number of slots: $Q = 12$
- Number of phases: $m = 3$ (A, B, C)
- Number of pole pairs: $p = 1$ (2 actual poles)
- Coil span: $y_q = 6$ slots (full-pitch winding)
- Number of turns: $N = 204$ turns
- Current amplitude: $I = 2.8$ A
- Electrical angle: $W_t = 90°$
- Layer type: Double layer

3.3.2 Step 1: Winding Matrix (6 × 12) - Double Layer

The winding matrix has 6 rows (3 phases × 2 layers):

```
Phase A top: [  0,   0,  +1,  +1,   0,   0,   0,   0,  -1,  -1,   0,   0
Phase B top: [ -1,  -1,   0,   0,   0,   0,  +1,  +1,   0,   0,   0,   0
Phase C top: [  0,   0,   0,   0,  -1,  -1,   0,   0,   0,   0,  +1,  +1
Phase A bot: [  0,   0,  +1,  +1,   0,   0,   0,   0,  -1,  -1,   0,   0
Phase B bot: [ -1,  -1,   0,   0,   0,   0,  +1,  +1,   0,   0,   0,   0
Phase C bot: [  0,   0,   0,   0,  -1,  -1,   0,   0,   0,   0,  +1,  +1
```

Slot numbers: 0 1 2 3 4 5 6 7 8 9 10 11

Summed Matrix (Top + Bottom for each phase)

```
Phase A sum: [  0,   0,  +2,  +2,   0,   0,   0,   0,  -2,  -2,   0,   0
Phase B sum: [ -2,  -2,   0,   0,   0,   0,  +2,  +2,   0,   0,   0,   0
Phase C sum: [  0,   0,   0,   0,  -2,  -2,   0,   0,   0,   0,  +2,  +2
```

3.3.3 Step 2: Calculate Instantaneous Currents

At $W_t = 90°$:

$$I_A = 2.8 \times \cos(90° - 0°) = 2.8 \times \cos(90°) = 2.8 \times 0 = 0 \text{ A}$$
$$I_B = 2.8 \times \cos(90° - 120°) = 2.8 \times \cos(-30°) = 2.8 \times 0.866 = 2.425 \text{ A}$$
$$I_C = 2.8 \times \cos(90° - 240°) = 2.8 \times \cos(-150°) = 2.8 \times (-0.866) = -2.425 \text{ A}$$

3.3.4 Step 3: Calculate Slot MMF for Each Phase

Formula: $\text{MMF}_{\text{slot}} = \text{Matrix}_{\text{sum}} \times I \times (N/2) = \text{Matrix}_{\text{sum}} \times I \times 102$

Phase A ($I_A = 0$)

Slot MMF_A: [0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0, 0] A-turns

Phase B ($I_B = 2.425$ A)

Slot MMF_B: [-495, -495, 0, 0, 0, 0, 495, 495, 0, 0, 0, 0] A-turns
(e.g., slot 0: $-2 \times 2.425 \times 102 = -495$ A-turns)

Phase C ($I_C = -2.425$ A)

Slot MMF_C: [0, 0, 0, 0, 495, 495, 0, 0, 0, 0, -495, -495] A-turns
(e.g., slot 4: $-2 \times (-2.425) \times 102 = 495$ A-turns)

Total Slot Current (sum of all phases)

Slot Total: [-495, -495, 0, 0, 495, 495, 495, 495, 0, 0, -495, -495] A-turns

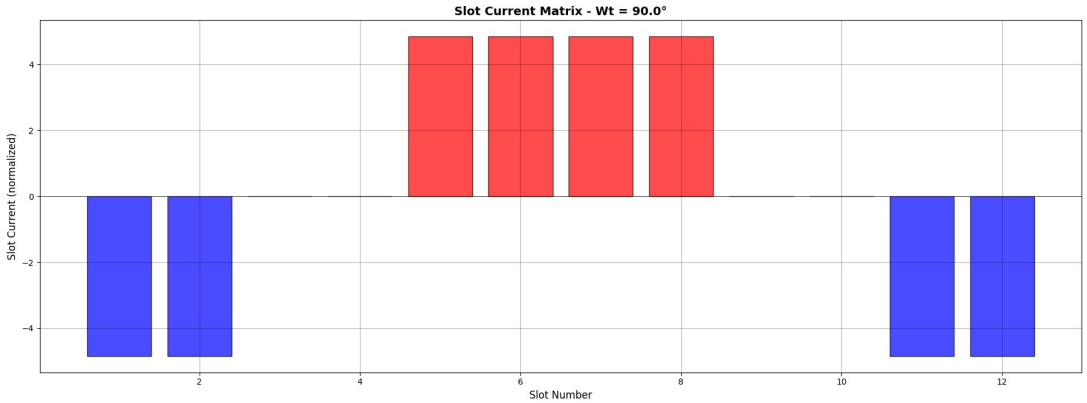


Figure 3.1: Slots current at $W_t = 90°$

3.3.5 Step 4: Calculate Tooth MMF (Iterative)

Starting from first non-zero slot (slot 0):

Initial Setup

- First non-zero slot: slot 0, value = −495 A-turns
- Count same value: 4 slots (checking circularly: slots 10, 11, 0, 1 all have −495 or −495)
- Left count for slot 0: 2 (slots 11 and 10 have same value)

Tooth MMF Calculations

Tooth 0 (initial): $\text{MMF}_{\text{tooth}}[0] = -4 \times (-495) + 2 \times (-495) \times (2+1) = 1980 - 2970 = -990$ A-turns

Tooth 1: $\text{MMF}_{\text{tooth}}[1] = \text{MMF}_{\text{tooth}}[0] + 2 \times \text{slot}[1] = -990 + 2 \times (-495) = -1980$ A-turns

Tooth 2: $\text{MMF}_{\text{tooth}}[2] = \text{MMF}_{\text{tooth}}[1] + 2 \times \text{slot}[2] = -1980 + 2 \times 0 = -1980$ A-turns

Tooth 3: $\text{MMF}_{\text{tooth}}[3] = \text{MMF}_{\text{tooth}}[2] + 2 \times \text{slot}[3] = -1980 + 2 \times 0 = -1980$ A-turns

Tooth 4: $\text{MMF}_{\text{tooth}}[4] = \text{MMF}_{\text{tooth}}[3] + 2 \times \text{slot}[4] = -1980 + 2 \times 495 = -990$ A-turns

Tooth 5: $\text{MMF}_{\text{tooth}}[5] = \text{MMF}_{\text{tooth}}[4] + 2 \times \text{slot}[5] = -990 + 2 \times 495 = 0$ A-turns

Tooth 6: $\text{MMF}_{\text{tooth}}[6] = \text{MMF}_{\text{tooth}}[5] + 2 \times \text{slot}[6] = 0 + 2 \times 495 = 990$ A-turns

Tooth 7: $\text{MMF}_{\text{tooth}}[7] = \text{MMF}_{\text{tooth}}[6] + 2 \times \text{slot}[7] = 990 + 2 \times 495 = 1980$ A-turns

Tooth 8: $\text{MMF}_{\text{tooth}}[8] = \text{MMF}_{\text{tooth}}[7] + 2 \times \text{slot}[8] = 1980 + 2 \times 0 = 1980$ A-turns

Tooth 9: $\text{MMF}_{\text{tooth}}[9] = \text{MMF}_{\text{tooth}}[8] + 2 \times \text{slot}[9] = 1980 + 2 \times 0 = 1980$ A-turns

Tooth 10: $\text{MMF}_{\text{tooth}}[10] = \text{MMF}_{\text{tooth}}[9] + 2 \times \text{slot}[10] = 1980 + 2 \times (-495) = 990$ A-turns

Tooth 11: $\text{MMF}_{\text{tooth}}[11] = \text{MMF}_{\text{tooth}}[10] + 2 \times \text{slot}[11] = 990 + 2 \times (-495) = 0$ A-turns

Final Tooth MMF Array

Table 3.1: Tooth MMF Values for 12-Slot Example

Tooth	0	1	2	3	4	5	6	7	8	9	10	11
MMF (A-turns)	-990	-1980	-1980	-1980	-990	0	990	1980	1980	1980	990	0

Note: These values represent the RESULTANT MMF at $W_t = 90°$. The individual phase tooth MMF values would be calculated separately for Phase A, B, and C before being combined using the weighting formula described in Step 6.

3.3.6 Step 5: Generate Continuous Profile

$$\text{Slot}_{\text{arc}} = 360°/(2 \times 12) = 15°$$
$$\text{Tooth}_{\text{arc}} = 360°/12 - 15° = 30° - 15° = 15°$$

For each tooth (e.g., Tooth 0 at angle 0°):

- Tooth center angle: $0° \times (360°/12) = 0°$
- Tooth region: $-7.5°$ to $+7.5°$ (around $0°$) $\rightarrow$ MMF $= -990$ A-turns (constant)
- Slot region: $7.5°$ to $22.5°$ $\rightarrow$ Linear interpolation from -990 to -1980 A-turns

For Tooth 7 at angle $210°$ ($7 \times 30°$):

- Tooth region: $202.5°$ to $217.5°$ $\rightarrow$ MMF $= 1980$ A-turns (constant)
- Slot region: $217.5°$ to $232.5°$ $\rightarrow$ Linear interpolation from 1980 to 1980 A-turns

This process creates 3600 points covering $0°$ to $360°$ with smooth transitions between teeth. The resulting waveform shows the characteristic stepped pattern of slotted machine MMF.

3.3.7 Step 6: Calculate Resultant MMF

At $W_t = 90°$, weighting factors:

$$\begin{aligned} \sin(90°) &= 1.0 \\ \sin(90° + 120°) &= \sin(210°) = -0.5 \\ \sin(90° - 120°) &= \sin(-30°) = -0.5 \end{aligned}$$

For each angle θ in the continuous profile:

$$\text{MMF}_{\text{resultant}}(\theta) = \text{MMF}_A(\theta) \times 1.0 + \text{MMF}_B(\theta) \times (-0.5) + \text{MMF}_C(\theta) \times (-0.5)$$

Since at $W_t = 90°$, $I_A = 0$ A, Phase A contributes zero MMF. The resultant is primarily due to phases B and C:

$$\text{MMF}_{\text{resultant}}(\theta) = \text{MMF}_B(\theta) \times (-0.5) + \text{MMF}_C(\theta) \times (-0.5)$$

This creates the stepped rotating wave pattern shown in the MMF plot with values ranging from -1980 to $+1980$ A-turns around the air gap. The pattern will rotate as W_t changes from $0°$ to $360°$, creating a traveling wave around the stator.

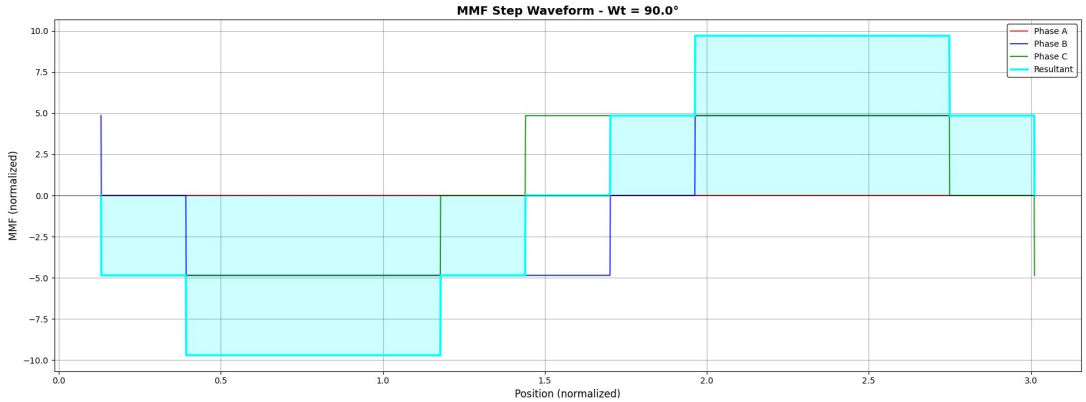


Figure 3.2: Resultant MMF with each phase contribution

3.3.8 Step 7: Harmonic Analysis

Apply DFT to the 12 tooth MMF values $[-990, -1980, -1980, -1980, -990, 0, 990, 1980, 1980, 1980, 990,$

For $\nu = 1$ (fundamental):

$$a_1 = \frac{2}{12} \times \sum_{j=0}^{11} [\text{MMF}_{\text{tooth}}[j] \times \cos(1 \times j \times 30^\circ)]$$

$$b_1 = \frac{2}{12} \times \sum_{j=0}^{11} [\text{MMF}_{\text{tooth}}[j] \times \sin(1 \times j \times 30^\circ)]$$

Calculating:

$$a_1 = \frac{2}{12} \times [-990 \times \cos(0^\circ) + (-1980) \times \cos(30^\circ) + (-1980) \times \cos(60^\circ) + \cdots + 0 \times \cos(330^\circ)]$$

$$b_1 = \frac{2}{12} \times [-990 \times \sin(0^\circ) + (-1980) \times \sin(30^\circ) + (-1980) \times \sin(60^\circ) + \cdots + 0 \times \sin(330^\circ)]$$

Amplitude: $A_1 = \sqrt{a_1^2 + b_1^2} \rightarrow$ This is the fundamental component

Phase angle: $\phi_1 = \text{atan2}(b_1, a_1)$

For $\nu = 5$ (5th harmonic):

$$a_5 = \frac{2}{12} \times \sum_{j=0}^{11} [\text{MMF}_{\text{tooth}}[j] \times \cos(5 \times j \times 30^\circ)]$$

$$b_5 = \frac{2}{12} \times \sum_{j=0}^{11} [\text{MMF}_{\text{tooth}}[j] \times \sin(5 \times j \times 30^\circ)]$$

Amplitude: $A_5 = \sqrt{a_5^2 + b_5^2}$

Percentage: $(A_5/A_1) \times 100\%$

For $\nu = 7, 11, 13$, etc., similar calculations yield the harmonic spectrum.

Typical results for this double-layer, full-pitch winding:

- Strong fundamental ($\nu = 1$)
- Presence of 5th, 7th, 11th, 13th harmonics
- Lower harmonic content than single-layer designs
- The harmonic spectrum reveals winding quality and performance characteristics

The waveform's shape directly corresponds to the harmonic content:

- Smooth sinusoidal $\approx$ pure fundamental
- Stepped/trapezoidal $\approx$ significant harmonics
- This 12-slot, 1 pole-pair configuration shows typical concentrated winding harmonics

3.4 Advantages of This Method

- Accurate representation of stepped MMF distribution in slotted machines
- Proper handling of both single and double-layer windings
- Phase-independent: automatically adapts to any number of phases (3, 5, 6, 7, etc.)
- Physically meaningful: respects Ampère's circuital law around the air gap
- Efficient computation: iterative method avoids complex integrations
- Suitable for fractional-slot concentrated windings and distributed windings
- Provides both spatial distribution and harmonic content

Chapter 4

Conclusion and Summary

4.1 Methodology Overview

This work has presented a comprehensive methodology for the design, analysis, and implementation of electrical machine windings, with particular emphasis on the calculation of Magnetomotive Force (MMF) distribution. The developed approach encompasses:

1. **Winding Generation**: Star of slots algorithm for balanced winding patterns for any phases/slots/poles configuration.
2. **MMF Calculation**: Tooth-centered iterative approach respecting Ampère's law.
3. **Harmonic Analysis**: DFT-based quantification of winding quality and performance.

4.2 Key Contributions

- **Universal Algorithm**: Handles any number of phases (2+) but preferably odd number of phases, slots, poles, and both integer/fractional slot windings.
- **Computational Efficiency**: Simple iterative method avoids complex integration while maintaining accuracy.
- **Practical Implementation**: Automatic feasibility checks, visualization tools, and performance prediction.

4.3 Applications

- **Design Optimization**: Rapid exploration of winding topologies and performance trade-offs.
- **Educational Tool**: Clear connection between theory and practical implementation.
- **Research Framework**: Support for novel winding topologies and harmonic mitigation strategies.

4.4 Key Insights

1. The star of slots provides a unified framework for diverse winding configurations.
2. Tooth-centered MMF calculation balances accuracy with computational simplicity.
3. Fractional slot windings ($q < 1$) offer unique advantages (compactness, fault tolerance) despite harmonic challenges.
4. Winding design requires holistic consideration of electromagnetic, thermal, and manufacturing factors.

4.5 Final Remarks

The presented methodology bridges theoretical electromagnetic principles with practical engineering implementation. Its universal nature ensures relevance across diverse applications—from industrial drives to electric vehicles and renewable energy systems. While providing sophisticated analytical tools, successful machine design ultimately requires engineering judgment and consideration of application-specific requirements.

This work contributes a versatile foundation for electric machine winding design, enabling systematic exploration of design spaces and informed decision-making in the complex optimization process that characterizes modern electric machine development.